

Government Reputation, FDI, and Profit Shifting*

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Abstract

I document that countries with higher investment risk have higher statutory corporate tax rates and that foreign direct investment inflows decline as investment risk rises. I rationalize these patterns in a static model in which multinational firms invest before knowing whether a revenue-maximizing government will implement its announced tax rate. Government reputation is the probability firms assign to a commitment type that must implement the announced rate. Firms can protect returns after a tax increase by shifting reported profits abroad once the implemented rate is known, while physical capital remains in place. Higher government reputation then makes the tax base more sensitive to the announced rate, inducing a lower tax. Without profit shifting, the optimal announced tax instead rises with government reputation. Extending the model to multiple periods, I show that future revenue losses can deter tax revisions, but the government may still announce a high tax that discourages investment. Profit shifting keeps investment positive at low government reputation because firms retain some profits after a tax increase. Higher government reputation increases the future revenue lost after a tax revision and can therefore support a lower announced tax and greater investment.

Keywords: Commitment, Corporate taxation, Foreign direct investment, Government reputation, Multinational firm, Profit shifting

JEL Classification: H21, H25, H26, F21, F23

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1 Introduction

The investment response to a low corporate tax rate depends on whether firms expect the government to implement that rate after they invest. A government may announce a low rate to attract capital and then raise the rate once that capital is installed. Uncertainty about whether the government will implement its announced tax rate is one source of investment risk. Government reputation matters because firms use past tax policy to assess whether the current announced tax rate will be implemented. Multinational firms can also protect part of their returns from a tax increase by shifting reported profits abroad. Profit shifting therefore affects both firms' willingness to invest and the revenue the government can obtain by changing its tax. I ask how profit shifting changes the optimal announced tax and the investment it attracts at different levels of government reputation.

Changes in taxation after investment can overturn the terms on which firms relied. In 2008, Zambia replaced the fiscal terms in existing mining development agreements and raised the corporate income tax rate from 25 to 30 percent ([International Monetary Fund, 2008](#)). In 2012, India extended taxation to past offshore transfers of Indian assets ([Ministry of Finance, Government of India, 2012](#)). The first episode changed an agreed rate, while the second changed the treatment of completed transactions. Both illustrate why investment decisions depend on the durability of tax policy.

I measure investment risk using the Investment Profile of the International Country Risk Guide (ICRG), which records risks to contracts, assets, and the receipt of investment returns. Countries with higher risk have higher statutory corporate tax rates, and foreign direct investment (FDI) inflows decline as investment risk rises. A synthetic control analysis of India's retrospective tax change shows a higher risk and lower inflows relative to the synthetic controls after the announcement. A change in the treatment of past investment can affect firms' expectations about future tax policy. This possibility motivates a model in which government reputation links past tax choices to the returns firms expect from new investment.

I study a model in which multinational firms allocate capital between a host country and a foreign location. The government of the host country maximizes corporate tax revenue. A commitment type must implement the tax announced before investment, whereas an opportunistic type can change that rate after firms invest. I refer to changing the announced tax rate after investment as a *tax revision*. Government reputation is modeled as the probability that firms assign to the commitment type. I study pooling equilibria in which both types announce the same tax and select the common tax announcement preferred by the commitment type. Firms remain uncertain about

government type when investing. After observing the implemented tax, firms can shift reported profits abroad while leaving physical capital in place.

I first study a static model in which the opportunistic type has no future revenue to protect. With profit shifting, firms retain some profits even after a tax revision. Higher government reputation increases the share of expected returns that depend on the announced rate. Under the model's restrictions on technology and profit-shifting costs, the tax base becomes more elastic, and the commitment type lowers its announced tax. Without profit shifting, higher government reputation instead raises investment without increasing this share. The larger capital stock makes the tax base less elastic and induces a higher announced rate. Profit shifting, therefore, reverses how the government adjusts its announced tax to an improvement in government reputation. In both economies, the expected implemented tax falls and investment rises.

I then develop a two-period model to ask whether future investment can eliminate the investment cost of uncertain government commitment. Government type remains fixed, and firms observe the first-period implemented tax before investing in the second period. A tax revision reveals the opportunistic type and raises the tax firms expect to face next period. Firms invest less, so the government loses future revenue. The opportunistic type can therefore prefer to implement the announced rate, even though it could raise the tax after investment.

Under full commitment, firms know that the government must implement any tax it announces. I show that investment can remain below this benchmark even when both types implement the announced tax. I refer to an announced tax that both types are willing to implement as a sustainable tax. Among the rates that both types implement, the government chooses a tax above the full-commitment rate when that benchmark leaves the opportunistic type with an incentive to raise the tax after investment. This higher rate reduces the extra revenue from a further increase but also lowers firms' returns. Hence, uncertainty about the government's commitment to the tax reduces investment through the chosen tax even when no tax revision occurs. In both economies, a higher government reputation raises future revenue lost after a tax revision and permits a lower announced tax while the incentive to raise it constrains the government's choice.

Profit shifting affects this tax choice through both current and future revenue. Firms' profit transfers limit the additional revenue from raising the tax on installed capital. Their ability to retain returns after a tax revision also leaves investment and revenue positive in the following period. I use a numerical comparison of economies with different profit-shifting costs to examine how the government adjusts its tax when these incentives change. Higher costs raise the current gain from revising the original tax more than the discounted future revenue loss. The government consequently

announces a higher rate, adding to the investment decline caused by higher costs at a fixed tax. This tax adjustment is larger with low government reputation than under full commitment. The comparison shows why the investment response to profit-shifting costs depends on the government's ability to implement a low tax and on firms' after-tax returns at a given rate.

Finally, I extend the model to multiple periods and characterize sustainable plans for taxes and firms' investment. Under these plans, government policies and firms' choices are optimal after every public tax history. The extension asks whether a longer investment relationship can sustain investment at a low government reputation. I first study plans in which both types implement announced taxes before the final period. I then examine the broader set of pooling equilibria, allowing for tax revisions and learning about government type. Without profit shifting, investment vanishes as government reputation approaches zero even in this broader set of equilibria. With profit shifting, firms retain some profits under any implemented tax and continue investing. Repeated investment, therefore, cannot fully replace firms' ability to protect returns in this model. Higher government reputation can still attract additional capital by permitting a lower tax that both types implement.

Literature Review I contribute to three strands of literature. First, I extend research on capital taxation under limited commitment, which studies how future investment restrains taxation of installed capital (Chari et al., 1988; Thomas and Worrall, 1994; Aguiar and Amador, 2011; Schnitzer, 1999). Chari and Kehoe (1990) formulate sustainable plans, and Phelan and Stacchetti (2001) characterize the equilibrium value set of a Ramsey tax game with capital accumulation and no uncertainty about government type. Ortigueira and Pereira (2022) show that retroactive income taxation can generate multiple equilibria driven by expectations in a neoclassical economy. Janeba (2000, 2002) study how multinational firms' ability to produce in more than one country constrains governments that lack commitment. In my model, firms protect returns by reporting profits abroad after physical capital is installed. This protection leaves investment and tax revenue positive even after the opportunistic type is revealed, changing the future revenue cost of a tax revision. In the model with multiple periods, profit shifting also sustains investment across pooling equilibria as government reputation approaches zero. Future investment alone cannot sustain positive investment in this limit without profit shifting.

Second, I contribute to the literature on government reputation. Backus and Driffill (1985) and King et al. (2008) study how learning about policymakers shapes monetary policy and the credibility of policy announcements. Phelan (2006) links tax policy and public trust to the type of government that can change. DAVIS and Kirpalani

(2020) examine fiscal rules that governments may choose not to enforce, and [Dovis and Kirpalani \(2021\)](#) characterize rules that preserve uncertainty by giving the optimizing type an incentive to follow them. Closest to the tax announcement problem, [Lu \(2013\)](#) allows the commitment type to choose its policy in a taxation game with Bayesian learning. I follow this approach to selecting tax announcements. Profit shifting changes both the revenue gained by raising the tax after investment and the future revenue lost after the government type is revealed. I show how these incentives determine the tax that both types are willing to implement. A higher government reputation permits a lower sustainable tax while the implementation incentive compatibility constraint binds. Thus, uncertainty about government type can reduce investment through the chosen tax rate even when both types implement that rate.

Third, I contribute to research on profit shifting and international taxation. Recent empirical work documents the scale of profit shifting and its consequences for measured output, productivity, and factor shares ([Guvenen et al., 2022](#); [Tørsløv et al., 2023](#)). Macroeconomic models also study how profit shifting affects intangible investment and optimal corporate taxation ([Dyrda et al., 2024a,b](#)). I complement this work by examining how profit shifting affects tax choices when governments cannot commit to the tax announced before investment. Profit shifting changes how government reputation affects the tax base's response to an announced tax. It reverses the optimal announced tax response in the static model and keeps investment positive at low government reputation when firms invest repeatedly.

This mechanism also relates to theories of international tax planning and tax competition. [Hong and Smart \(2010\)](#) show how international tax planning can support high corporate taxes by protecting returns to mobile capital. Related work studies tax competition with profit shifting ([Peralta et al., 2006](#); [Devereux et al., 2008](#); [Johannesen, 2010](#); [Mongrain et al., 2023](#)). Closest to my static model, [Mardan and Stimmelmayer \(2020\)](#) obtain a reversal of the risk–tax relation with an exogenous probability of project failure. In my model, firms face the risk that the opportunistic type raises the tax after they invest. [de Mooij and Liu \(2020\)](#) find that transfer-pricing regulations reduce investment by multinational affiliates, with no significant decline in investment by the multinational group as a whole. My numerical comparison shows how higher profit-shifting costs can make the originally announced tax harder to sustain, inducing a higher rate and a larger investment decline than under full commitment.

Outline Section 2 presents the empirical evidence. Section 3 develops the model, and Section 4 derives the relationship between government reputation and optimal taxation in one period. Section 5 studies how future revenue affects the government's willingness to implement its announced tax. Section 6 characterizes sustainable taxes

and examines investment at low government reputation in a model with multiple periods. Section 7 concludes.

2 Investment Risk, Taxation, and Capital Inflows

I examine how investment risk is associated with corporate tax rates and subsequent investment inflows. The tax regressions ask whether countries with higher risk offer lower statutory rates. The investment regressions ask whether an increase in risk is followed by lower inflows. I then consider India's 2012 retrospective tax change to connect these patterns to a specific tax-policy event.

2.1 Investment Risk and Corporate Tax Rates

I measure investment risk using the Investment Profile of the International Country Risk Guide (ICRG). The index combines Contract Viability/Expropriation, Profits Repatriation, and Payment Delays. These components concern the security of firms' contracts and assets and their ability to receive investment returns. The combined score ranges from zero to twelve, with higher values indicating lower risk (The PRS Group, nd). For country h and year t , I define

$$\text{Risk}_{ht} = 12 - \text{Investment Profile}_{ht},$$

where Investment Profile is the annual average of the monthly score. Higher values of Risk variable indicate greater investment risk.

I use the Organization for Economic Co-operation and Development (OECD) measure of the statutory corporate income tax rate, which includes central and subnational taxation (OECD, 2026). I estimate the following linear regression specified as

$$\text{Tax rate}_{ht} = \alpha_h + \lambda_t + \beta_{\text{tax}} \text{Risk}_{h,t-1} + \delta' X_{ht} + \phi' Z_{h,t-1} + \varepsilon_{ht}. \quad (1)$$

The dependent variable is the statutory tax rate in percentage points. Country fixed effects α_h control for persistent differences across countries, and year fixed effects λ_t control for changes common to all countries. The coefficient of interest is β_{tax} . It measures the difference in the statutory tax rate associated with a one-point increase in the previous year's investment risk, conditional on the fixed effects and controls.

The economic controls X_{ht} are log real gross domestic product (GDP) per capita, log population, and natural-resource rents relative to GDP. The institutional and political controls $Z_{h,t-1}$ are the ICRG measures of corruption, law and order, bureaucratic quality, government stability, internal conflict, and external conflict. All ICRG measures are

expressed as risks and lagged one year. Table 1 adds the institutional and political controls separately and together.

Table 1 shows that higher investment risk is associated with higher statutory corporate tax rates. With economic, institutional, and political controls, a one-point increase in risk is associated with a 0.426 percentage-point higher rate (column (4)). Adding lagged real GDP growth leaves the result similar. The positive association also holds in the sensitivity checks reported in Appendix Table 4.

The tax measure records the rate specified in law. Tax terms announced before firms invest and subsequent tax revisions to those terms are not observed separately.

2.2 Investment Following Changes in Risk

I next examine whether an increase in investment risk is followed by lower direct-investment inflows. I use the quarterly inflows of [Forbes and Warnock \(2021\)](#), aggregated to annual observations. Log annual inflows are denoted by $f_{ht} = \log F_{ht}$, and the change in investment risk is $\Delta R_{ht} = \text{Risk}_{ht} - \text{Risk}_{h,t-1}$. Following [Jordà \(2005\)](#), I estimate a separate local projection by ordinary least squares for $s = 0, 1, 2, 3$ specified as

$$f_{h,t+s} - f_{h,t-1} = \alpha_h^s + \lambda_t^s + \beta_s \Delta R_{ht} + \gamma_s \Delta R_{h,t-1} + \delta_s \text{Risk}_{h,t-1} + \phi_s \Delta f_{h,t-1} + \eta_s' X_{h,t-1} + \psi_s' Z_{h,t-1} + \varepsilon_{ht}^s. \quad (2)$$

The dependent variable is the change in log inflows from the year before the risk change to year $t + s$. Each regression includes country fixed effects α_h^s and year fixed effects λ_t^s . The coefficient of interest, β_s , measures the change in log annual inflows associated with a one-point increase in investment risk in year t . The remaining regressors are the change in risk, the risk level, inflow growth, and the economic, institutional, and political controls defined above, all lagged one year.

Figure 1 illustrates the estimated response β_s of FDI inflows to a one-point increase in investment risk.¹ A one-point increase in risk is associated with a 0.243 log-point decline in annual inflows from the year before the risk change to the following year. The coefficient remains negative at horizon two and is close to zero at horizon three.

2.3 India's 2012 Retrospective Tax Change

The panel regressions concern changes in a broad measure of investment risk. To examine a specific change in tax policy, I turn to India's 2012 retrospective tax legislation. On

¹Appendix Figure 6 adds institutional and political controls and shows a similar response. Appendix A.3 examines alternative measures of inflows.

TABLE 1: Investment risk and statutory corporate tax rates

	(1)	(2)	(3)	(4)	(5)
Investment risk	0.395*** (0.148)	0.394*** (0.149)	0.435*** (0.142)	0.426*** (0.143)	0.420*** (0.145)
Economic controls	Yes	Yes	Yes	Yes	Yes
Institutional controls	No	Yes	No	Yes	Yes
Political controls	No	No	Yes	Yes	Yes
Lagged GDP growth	No	No	No	No	Yes
Country and year effects	Yes	Yes	Yes	Yes	Yes
Observations	1,887	1,887	1,887	1,887	1,887
Countries	101	101	101	101	101
Within R^2	0.025	0.031	0.033	0.038	0.039

Notes: OECD annual rates over 2003–2021, with 1,887 observations in 101 countries. Zero-rate observations in Haiti and Togo are excluded as described in Appendix Table 4, which reports estimates with and without these observations. All columns include country and year effects. Economic controls are log real GDP per capita, log population, and resource rents/GDP. Institutional controls are corruption, law and order, and bureaucratic quality. Political controls are government stability, internal conflict, and external conflict. All ICRG controls are lagged risks. Parentheses contain country-clustered standard errors. ***, **, and * denote significance at the 1, 5, and 10 percent levels. Appendix Table 3 gives all slope coefficients for columns (4) and (5).

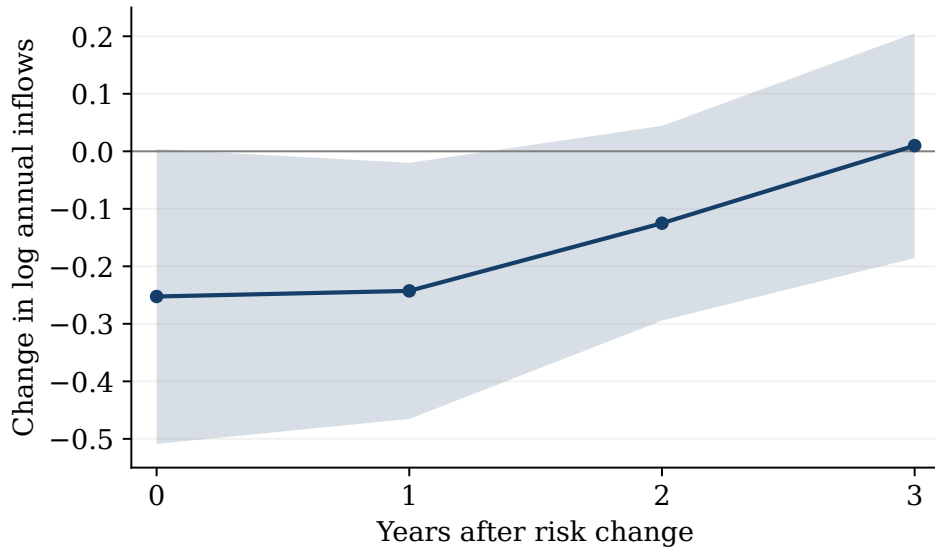


FIGURE 1: FDI Inflows after an Increase in Investment Risk

Notes: The line plots β_s from (2) with economic controls. The estimates measure the change in log annual inflows between $t - 1$ and $t + s$ associated with a one-point increase in risk in year t . The sample has 628 observations in 55 countries, with risk changes dated 2003–2016 and outcomes through 2019. Log outcomes require positive inflows at all dates used in the regressions. Annual inflows require four quarterly observations. Controls include country and year effects, lagged risk level and change, lagged inflow growth, and lagged log real GDP per capita, log population, and resource rents/GDP. The shaded band is a 95 percent confidence interval using country-clustered standard errors.

March 16, the Finance Bill proposed extending taxation to completed offshore trans-

fers of Indian assets (Ministry of Finance, Government of India, 2012). The proposal followed the January Supreme Court ruling in the Vodafone dispute and became law in May (Ministry of Finance, Government of India, 2021). It changed the treatment of past transactions rather than the headline corporate tax rate.

I use the synthetic-control method of Abadie et al. (2010) to compare India’s risk and direct-investment inflows with a weighted average of other countries that form the donor pool. The weights are chosen separately for each outcome to match India’s path before the tax announcement and are then held fixed. I refer to the resulting weighted average as the synthetic comparison. Inflows are divided by each country’s pre-event mean to account for differences in size. Appendix A.4 describes the method, reports country weights, and examines sensitivity.

Figure 2 shows that India’s investment risk rises and FDI inflows decline relative to their synthetic comparisons after the tax announcement. Average inflows remain slightly above India’s pre-event mean but grow less than in the synthetic comparison.

Investment risk rose in February before the March tax announcement and increased further in May. The increase before the announcement may reflect anticipation of the policy change. A general anti-avoidance proposal and a broader slowdown associated with policy uncertainty and weaker business confidence also occurred during this period (Anand and Tulin, 2014). The comparison describes investment conditions around the tax announcement when these other policies and economic conditions were also changing.

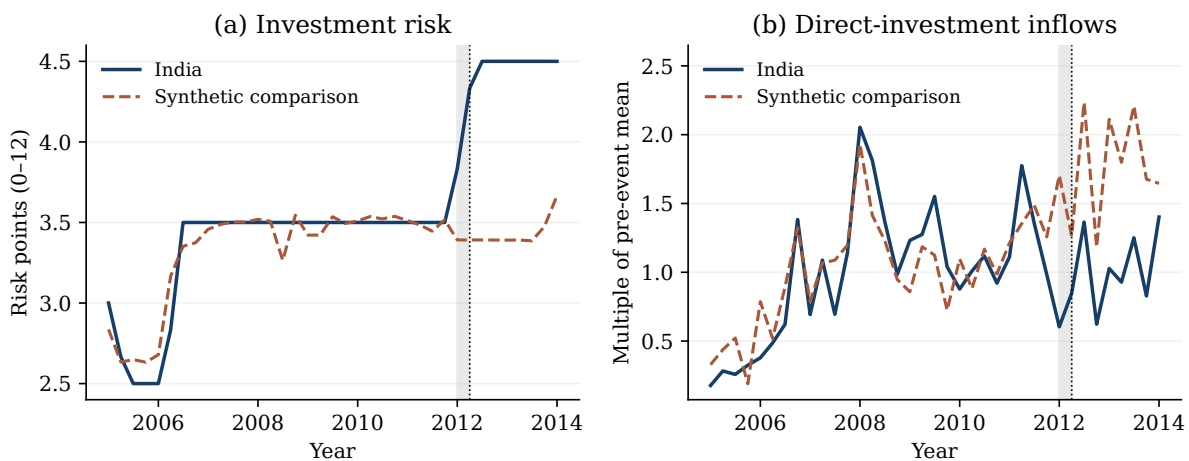


FIGURE 2: Investment risk and inflows around India’s 2012 tax announcement

Notes: The donor pool comprises 57 countries with complete quarterly risk and inflow observations over 2005Q1–2014Q1 and positive pre-event mean inflows. Separate weights for risk and inflows are fitted over 2005Q1–2011Q4; the comparison after the tax announcement is 2012Q2–2014Q1. Inflows are divided by each country’s pre-event mean. The shaded quarter contains the January ruling and March tax announcement and is excluded from fitting the weights; the dotted line marks the following quarter. Appendix Table 5 reports all donor weights.

The evidence concerns the security of investment returns. Changes in the tax treatment of existing investments can also affect firms' expectations about how returns on new investments will be taxed.

3 Model

I begin with a one-period economy in which multinational firms can shift reported profits across countries. Firms invest before learning whether the government will implement its announced tax rate. This timing captures the exposure of installed capital to subsequent tax choices. Policy credibility is one determinant of investment conditions emphasized by [World Bank \(2004, pp. 45–50\)](#).

A commitment type must implement the tax announced before investment, whereas an opportunistic type can revise the tax after firms invest. Firms do not observe the type of government. Government reputation p is the probability firms assign to the commitment type. Firms choose investment using the expected after-tax returns under the two types. Their belief about commitment, therefore, affects the risk of a tax revision and the returns they expect. The model studies this tax-policy source of investment risk. The ICRG index covers risks to contracts and payments more broadly and therefore does not directly measure p . The model concerns capital allocated to production, whereas the empirical FDI measure records financial inflows.

3.1 Environment and Timing

There is a unit measure of identical foreign-owned multinational firms. Each firm has access to a fixed amount of capital $\bar{k} > 0$ and allocates it between the host country and a foreign location. Capital employed in the host country, denoted by k , produces output zk^α , where $z > 0$ and $0 < \alpha < 1$. The rental cost per unit of capital is $r^* \geq 0$ and is deductible where capital is employed. Operating profits are therefore $\rho(k) = zk^\alpha - r^*k$ in the host country and $\rho(\bar{k} - k)$ abroad. The host country is a small open economy that takes the foreign corporate tax rate $\tau^* \in (0, 1)$ as given.

Both types of government maximize corporate tax revenue. Each type applies a uniform tax rate to all firms. The types differ only in whether they can revise the announced tax after investment.

Before investing, firms know the current tax rules, deductible costs, reporting opportunities, production technology, and public policy history. Each firm can operate and shift profits across both locations. The prior p summarizes firms' information about government type. Firms are uncertain whether the government will implement its announced tax after capital is installed.

The government announces a tax τ^R before firms invest and implements a rate $\tau \in [0, 1]$ afterward. Firms then choose a profit transfer θ . A positive transfer moves reported profits into the host country, and a negative transfer moves profits abroad. All tax announcements and implemented rates are public. Figure 3 summarizes the timing.

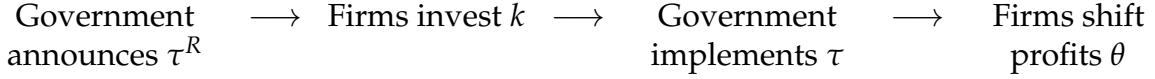


FIGURE 3: Timing within a period

Installed capital is fixed when the government implements the tax. I assume that firms can adjust profit transfers within an existing multinational structure before filing taxes. [Hopland et al. \(2018\)](#) find within-year flexibility in transfer pricing, but not in internal debt, in response to operating losses. The model holds firms' ability to shift profits fixed and allows the amount transferred to respond to the implemented tax. Firms anticipate the transfers and their resource costs when investing.

3.2 Firms' Profit-Shifting Choice

Profit shifting incurs a convex resource cost in the country from which profits are transferred. For an outward transfer, the affiliate in the host country incurs $C_H(k, \theta) = \gamma\theta^2/[2\rho(k)]$, and the foreign affiliate incurs no cost, i.e., $C_F(k, \theta) = 0$. For an inward transfer, $C_H(k, \theta) = 0$ and $C_F(k, \theta) = \gamma\theta^2/[2\rho(\bar{k} - k)]$. Both costs are zero when $\theta = 0$. The parameter $\gamma > 0$ measures the cost of shifting profits. Each affiliate deducts these costs from its taxable profits, and each country's tax base must be nonnegative. An affiliate with no operating profits cannot transfer profits to the other country.

Given the capital and the implemented tax, the firm solves

$$\begin{aligned}
 \max_{\theta} \quad & (1 - \tau)[\rho(k) + \theta - C_H(k, \theta)] \\
 & + (1 - \tau^*)[\rho(\bar{k} - k) - \theta - C_F(k, \theta)], \\
 \text{subject to} \quad & \rho(k) + \theta - C_H(k, \theta) \geq 0, \\
 & \rho(\bar{k} - k) - \theta - C_F(k, \theta) \geq 0.
 \end{aligned} \tag{3}$$

The largest share of operating profits that can be transferred without making the transferring affiliate's tax base negative is $\bar{y} = 2/(\sqrt{1 + 2\gamma} + 1)$. The optimal transfer is

$$\theta(k, \tau) = \begin{cases} -\rho(k) \min\left\{\frac{\tau - \tau^*}{\gamma(1 - \tau)}, \bar{y}\right\}, & \tau^* \leq \tau < 1, \\ \rho(\bar{k} - k) \min\left\{\frac{\tau^* - \tau}{\gamma(1 - \tau^*)}, \bar{y}\right\}, & \tau < \tau^*. \end{cases} \tag{4}$$

At $\tau = 1$, the share of profits transferred abroad equals $\bar{y}$. A tax above the foreign rate induces outward profit shifting, while a tax below the foreign rate induces inward profit shifting.

For outward transfers, $y(\tau) = -\theta(k, \tau)/\rho(k)$ is the share of profits earned in the host country that the firm reports abroad. The firm's after-tax return and the government's tax revenue per unit of these profits are

$$\begin{aligned} A(\tau) &= (1 - \tau) \left[1 - y(\tau) - \frac{\gamma y(\tau)^2}{2} \right] + (1 - \tau^*) y(\tau), \\ v(\tau) &= \tau \left[1 - y(\tau) - \frac{\gamma y(\tau)^2}{2} \right]. \end{aligned} \quad (5)$$

The firm retains $A(\tau)\rho(k) + (1 - \tau^*)\rho(\bar{k} - k)$ after taxes and profit-shifting costs. At a tax rate of one, $A(1) = (1 - \tau^*)\bar{y} > 0$ because the firm can report profits abroad.

For the tax comparisons below, I impose the following restrictions on production and profit-shifting costs

$$\begin{aligned} \alpha z \bar{k}^{\alpha-1} &> (2 - \alpha)r^*, \\ \gamma &> \frac{\tau^*}{1 - \tau^*} + \frac{(\tau^*)^2}{2(1 - \tau^*)^2}, \\ \frac{\tau^*}{\gamma(1 - \tau^*)} \left(\frac{\sqrt{1 + 2\gamma} + 1}{2} \right)^{\alpha/(1-\alpha)} + \frac{\tau^* \alpha}{(1 - \alpha)(1 - \tau^*)} &< 1, \\ \frac{\sqrt{1 + 2\gamma} - 1}{2} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right] &< 1 - \alpha. \end{aligned} \quad (6)$$

The production restriction ensures that marginal profits are positive at every feasible capital allocation. The next two restrictions imply outward profit shifting at the government's optimal tax rates, derived below. The last restriction limits how the responsiveness of investment changes with expected returns and is used to determine how government reputation affects the optimal announced tax.

3.3 Taxation after Investment

When all firms choose the same capital allocation, the unit measure of firms implies aggregate capital $K = k$ in the host country. Government revenue at tax τ is

$$\mathcal{R}(K, \tau) = \tau[\rho(K) + \theta(K, \tau) - C_H(K, \theta(K, \tau))].$$

The commitment type implements τ^R . The opportunistic type chooses

$$\tau^O(K) \in \arg \max_{\tau \in [0,1]} \mathcal{R}(K, \tau). \quad (7)$$

I refer to the rate that maximizes revenue after investment as the discretionary tax. The opportunistic type takes installed capital as given but anticipates firms' profit-shifting choices. A higher tax raises revenue per unit of reported profit and reduces the reported tax base through profit shifting.

3.4 Firms' Investment Choice

Firms invest after observing the tax announcement and before the implementation of the tax. When both government types announce the same rate, the tax announcement conveys no information about the type. Firms expect τ^R with probability p and $\tau^O(K)$ with probability $1 - p$. Each firm chooses

$$\begin{aligned} \max_{k \in [0, \bar{k}]} \mathbb{E}_\tau \Big\{ & (1 - \tau)[\rho(k) + \theta(k, \tau) - C_H(k, \theta(k, \tau))] \\ & + (1 - \tau^*)[\rho(\bar{k} - k) - \theta(k, \tau) - C_F(k, \theta(k, \tau))] \Big\}. \end{aligned} \quad (8)$$

Each firm takes the implemented taxes as given and does not internalize the effect of aggregate investment on the government's tax choice. Investment and discretionary taxation must jointly satisfy individual optimality, $k = K$, and (7). I denote the resulting investment in the host country by $K(\tau^R, p)$.

When both implemented taxes exceed the foreign rate, the investment condition is

$$[pA(\tau^R) + (1 - p)A(\tau^O)]\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (9)$$

The expected marginal return on investment in the host country equals the marginal return forgone abroad. Profit shifting affects investment through the after-tax return retained under each type of government.

3.5 Tax Announcements

In a pooling equilibrium, both types make the same tax announcement. Following [Lu \(2013\)](#), I select the tax announcement preferred by the commitment type. Each government type must prefer this tax announcement to every alternative. I refer to these requirements as the announcement incentive compatibility constraints. A tax

announcement different from the prescribed rate is off path and leads firms to assign a probability of one to the opportunistic type.

Investment, when the government is known to be the opportunistic type, is denoted by K_0 . Revenue in that equilibrium is

$$R_0 = \max_{\tau \in [0,1]} \mathcal{R}(K_0, \tau).$$

I use this equilibrium after every off-path tax announcement. Firms then invest K_0 and either type can collect at most R_0 on that capital. The commitment type's preferred common tax announcement solves

$$\begin{aligned} \max_{\tau^R \in [0,1]} \quad & \mathcal{R}(K(\tau^R, p), \tau^R), \\ \text{subject to} \quad & \mathcal{R}(K(\tau^R, p), \tau^R) \geq R_0, \\ & \mathcal{R}(K(\tau^R, p), \tau^O(K(\tau^R, p))) \geq R_0. \end{aligned} \tag{10}$$

The first constraint requires the commitment type to weakly prefer the common tax announcement over any off-path tax announcement. The second imposes the same requirement on the opportunistic type, which optimizes the tax after investment. Appendix B.1 proves the existence of the selected pooling equilibrium and justifies the selection criterion.

I refer to the commitment type's preferred common tax announcement as the optimal announced tax and denote it by $\tau^R(p)$. The choice takes firms' uncertainty about government type into account. I compare pooling equilibria at the same prior p . The off-path belief described above applies to deviations within each equilibrium. Full commitment corresponds to $p = 1$, when firms know that the government must implement its announced tax.

4 Government Reputation and Optimal Taxation

I first ask how firms' ability to protect returns changes the tax announced before investment. I compare the baseline economy with an otherwise identical economy in which firms cannot shift profits. Within each economy, I vary the prior probability p while holding technology, the foreign tax, and profit-shifting opportunities fixed. There is only one investment decision, so the government has no future tax revenue to lose after a tax revision.

In equilibrium, the opportunistic type chooses the same tax τ^O at every level of government reputation. The tax solves

$$(2\gamma + 1)(1 - \tau^O)^3 = (1 - \tau^*)^2(1 + \tau^O). \quad (11)$$

Under the cost restrictions in Section 3, the optimal discretionary tax exceeds the foreign tax and is below one. The assumed profit-shifting costs cause the reported tax base to decline as the tax rises and vanish at a sufficiently high rate. Hence, the government maximizes revenue before the tax base falls to zero. Firms retain a positive share $A(\tau^O) > 0$ of profits earned in the host country, so investment remains positive even when $p = 0$.

Proposition 1 (Government reputation lowers the optimal announced tax with profit shifting). *In the economy defined in Section 3, the optimal announced tax $\tau^R(p)$ is unique. It satisfies $\tau^* < \tau^R(p) < \tau^O$ for every $p \in (0, 1]$, and for every $0 < p < p' \leq 1$,*

$$\tau^R(p') < \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Proof. See Appendix B.1. □

At the tax announcement stage, the commitment type weighs the revenue from a higher rate against the reduction in its tax base. Raising the announced tax reduces investment and the share of profits reported in the host country. The investment response depends on how much of the firms' expected returns is affected by the announced tax. A marginal increase in that tax lowers returns only when the government is the commitment type. Firms also earn positive returns under discretionary taxation, so a higher government reputation increases the fraction of expected returns that are exposed to the announced rate. At a fixed announced tax, a marginal increase in that rate therefore causes a larger proportional reduction in expected returns.

Higher government reputation also raises investment at a given tax rate. With more capital installed, a proportional fall in expected returns reduces profits by a smaller percentage. Under the maintained restrictions, the greater sensitivity of expected returns to the announced tax outweighs this effect. The tax base therefore becomes more elastic with respect to that rate, and the commitment type chooses a lower tax.

The probability of a tax revision in this one-period equilibrium is $1 - p$. A higher government reputation lowers this risk and the announced tax, raising investment through both changes. The lower tax implemented by the commitment type also reduces the share of profits shifted abroad. The commitment type collects more revenue from the larger tax base despite the lower tax rate. The opportunistic type earns

more revenue at the same discretionary tax because profits earned in the host country increase.

Without profit shifting, firms cannot shield profits earned in the host country from taxation. The opportunistic type chooses $\tau^O = 1$. Firms retain part of the profits they earn in the host country only when the government is of the commitment type, so their investments satisfy

$$p(1 - \tau^R)\rho'(K) = (1 - \tau^*)\rho'(\bar{k} - K). \quad (12)$$

At a fixed announced tax, a higher government reputation raises expected returns and investment. Because firms retain profits earned in the host country only under the commitment type, a marginal tax increase reduces expected returns on investment in the host country by the same proportion at every level of government reputation. Higher investment reduces the elasticity of profits earned in the host country with respect to the after-tax return. A marginal tax increase, therefore, causes a smaller proportional contraction of the tax base, and the commitment type chooses a higher rate.

Proposition 2 (Government reputation raises the optimal announced tax without profit shifting). *In the economy without profit shifting, under the production assumptions in Section 3, (10) has a unique maximizer $\tau^R(p) \in (0, 1)$ for every $p \in (0, 1]$. For every $0 < p < p' \leq 1$,*

$$\tau^R(p') > \tau^R(p), \quad K(\tau^R(p'), p') > K(\tau^R(p), p).$$

Proof. See Appendix B.2. □

Propositions 1 and 2 show that profit shifting reverses the response of the optimal announced tax to higher government reputation. The same improvement in government reputation induces the government to choose a lower rate with profit shifting and a higher rate without it. To relate this comparison to the tax data, I also consider the tax rate firms expect to face. Without profit shifting, the announced and expected tax rates move in opposite directions. Although $\tau^R(p)$ rises with government reputation, Proposition 2 shows that investment also rises. The investment condition (12) then requires $p(1 - \tau^R)$ to increase, so the expected tax rate $1 - p(1 - \tau^R)$ falls. The decline in the probability of confiscation therefore outweighs the increase in the announced rate.

With profit shifting, the expected tax rate $p\tau^R(p) + (1 - p)\tau^O$ also falls with government reputation because both the announced tax and the probability of facing τ^O decline. A positive association between risk and observed statutory rates is therefore compatible with both economies. The propositions distinguish the response of the chosen tax announcement, which the annual tax data do not separately observe.

Figure 4 illustrates the tax and investment responses to government reputation. Appendix C.1 describes the numerical method.

In Figure 4, increasing p from 0.1 to 0.9 lowers τ^R from 34.3 to 30.8 percent. The discretionary tax and the share of profits shifted after a tax revision are unchanged. The amount shifted after a tax revision nevertheless rises with government reputation because firms invest more and earn higher profits in the host country.

5 Government Reputation and Future Investment

In the one-period economy, the opportunistic type revises an announced tax below τ^O because it has no future revenue to lose. When firms invest again, their response

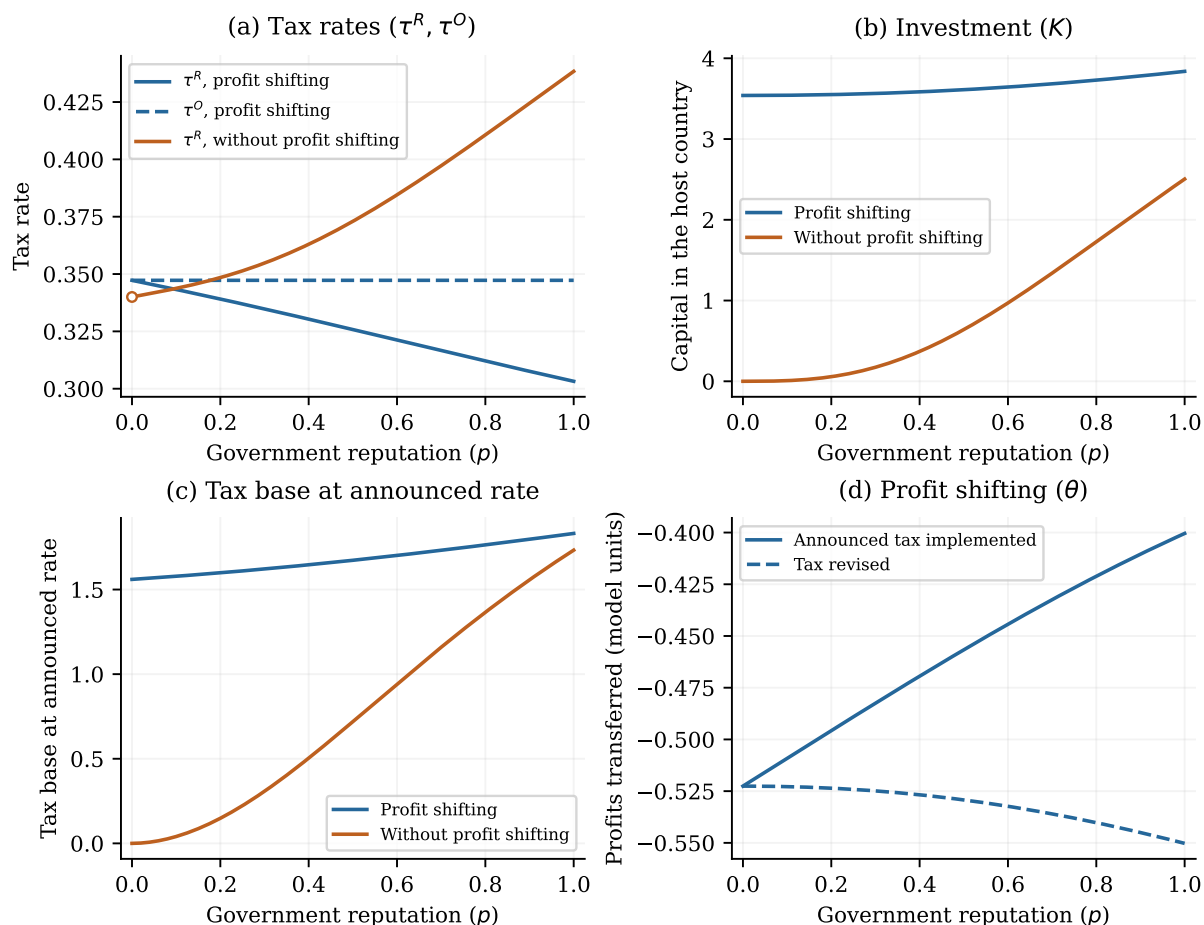


FIGURE 4: Government reputation, tax policy, investment, and profit shifting

Notes: The parameters are $z = 1$, $\alpha = 0.66$, $r^* = 0.04$, $\bar{k} = 10$, $\tau^* = 0.15$, and $\gamma = 1.25$, with the same production parameters in the comparison without profit shifting. Costs are incurred and deducted where profits originate. Panel (a) reports the optimal announced tax and the discretionary tax. The open circle denotes the limit of the announced tax as government reputation approaches zero without profit shifting. Panel (c) reports the tax base when the announced rate is implemented. Negative transfers in panel (d) denote outward profit shifting. The discretionary tax without profit shifting is one.

to a tax revision can make it profitable to implement the currently announced tax rate. I introduce a second period to determine whether this incentive also removes the investment cost of uncertain commitment. Profit shifting affects this incentive because firms can move reported profits after a tax increase and continue investing after the opportunistic type is revealed.

5.1 Two-Period Model

The same firms operate in both periods. Each firm rents $\bar{k}$ units of capital in each period and allocates them between the host country and the foreign location. Capital fully depreciates after production, so each period's capital stock equals that period's new investment. The rental cost r^* includes depreciation. Firms distribute their profits, and their access to capital in the next period is independent of current earnings. Firms retain their information when replacing capital, so past tax policy affects current investment through beliefs about government type.

Firms maximize expected discounted profits. An individual firm cannot affect aggregate capital or public beliefs, and its current capital does not survive into the next period. The firm therefore solves the one-period investment problem at each history. Government type remains fixed. The government maximizes the sum of current revenue and second-period revenue discounted by $\beta \in (0,1)$. Initial government reputation is $p \in (0,1)$.

Within each period, the government announces a tax rate, firms invest, the government sets the implemented rate, and firms choose their profit transfers. Firms observe the first-period tax announcement and the implemented tax before choosing their second-period investment. In the second period, there is no future investment to discipline tax policy. I use the one-period equilibrium selected in Section 3.5 as the terminal equilibrium.

The technology and profit-shifting costs are those in Section 3. With profit shifting, the discretionary tax is the constant τ^O derived in Section 4, and revenue under outward profit shifting is $\mathcal{R}(K, \tau) = v(\tau)\rho(K)$. The equilibrium at $p = 0$ gives $K_0 = K(\tau^O, 0)$ and $R_0 = \mathcal{R}(K_0, \tau^O)$. The commitment and opportunistic types' second-period revenues at government reputation q are $V_C(q)$ and $V_O(q)$, with $V_C(0) = V_O(0) = R_0$. I refer to these second-period revenues as continuation revenues.

Higher government reputation in the second period induces more investment in the selected one-period equilibrium. The opportunistic type then collects revenue at the fixed rate τ^O on higher profits earned in the host country, so $V_O(q)$ increases. Higher government reputation also raises the commitment type's revenue at a fixed announced tax. Its maximized revenue $V_C(q)$ therefore increases as well. Both continuation

revenues are also increasing in government reputation without profit shifting, with $\mathcal{R}(K, \tau) = \tau \rho(K)$, $\tau^O = 1$, and $R_0 = 0$.

5.2 Tax Implementation and Tax Announcements

For a candidate common first-period tax announcement τ , the commitment type implements τ , and the opportunistic type implements it with a probability of m . When $\tau \neq \tau^O$, the overall implementation probability is denoted by u , and the posterior probability of the commitment type after implementation is denoted by q . Bayes' rule gives

$$u = p + (1 - p)m, \quad q = \frac{p}{u}. \quad (13)$$

First-period investment is $K(\tau, u)$, using the investment function from the one-period model with u in place of p . Any tax revision reveals the opportunistic type and reduces government reputation to zero. The opportunistic type maximizes current revenue at τ^O whenever it revises the announced tax.

The probability of a tax revision is $1 - u = (1 - p)(1 - m)$. Firms' exposure to this risk depends on both their beliefs about the type of government and the incentives of the opportunistic type to implement the announced tax. Since $uq + (1 - u)0 = p$, firms' expected posterior equals the prior. Government reputation is therefore a martingale. Implementing the announced tax raises government reputation when $m < 1$, offset in expectation by the possible fall to zero after a tax revision. When $m = 1$, the risk of a tax revision is zero even though firms remain uncertain about the government type. Government reputation remains unchanged, but the revenue that would be lost after a tax revision deters the opportunistic type from revising the tax.

The opportunistic type compares the current gain from a tax revision with the discounted loss of future revenue. The net gain is

$$D(\tau, m; p) = \mathcal{R}(K(\tau, u), \tau^O) - \mathcal{R}(K(\tau, u), \tau) - \beta[V_O(q) - R_0]. \quad (14)$$

The opportunistic type implements the announced tax with certainty if $D \leq 0$ at $m = 1$ and revises the tax with certainty if $D \geq 0$ at $m = 0$. For $0 < m < 1$, the opportunistic type must be indifferent between implementing and revising the announced tax, so $D = 0$. I refer to these restrictions on the opportunistic type's decision as the implementation incentive compatibility conditions. They must hold jointly with firms' optimal investment choices and Bayes' rule for q . When $m = 0$, implementation of the announced tax identifies the commitment type, giving $q = 1$.

As in Section 3.5, I select the common tax announcement that maximizes the commitment type's discounted revenue, subject to both types' announcement incentive

compatibility constraints. Tax announcements above τ^O are dominated, as shown in Appendix B.3. The commitment type solves

$$\max_{\tau \in [0, \tau^O]} \{ \mathcal{R}(K(\tau, u(\tau)), \tau) + \beta V_C(q(\tau)) \}, \quad (15)$$

where $u(\tau)$ and $q(\tau)$ are determined by the implementation incentive compatibility conditions above. An off-path tax announcement leads firms to assign a probability of zero to the commitment type in both periods. Either type can earn at most $(1 + \beta)R_0$ after an off-path tax announcement.

The commitment type's and the opportunistic type's incentive compatibility constraints at the tax announcement stage are

$$\begin{aligned} \mathcal{R}(K(\tau, u), \tau) + \beta V_C(q) &\geq (1 + \beta)R_0, \\ m[\mathcal{R}(K(\tau, u), \tau) + \beta V_O(q)] + (1 - m)[\mathcal{R}(K(\tau, u), \tau^O) + \beta R_0] &\geq (1 + \beta)R_0. \end{aligned} \quad (16)$$

Appendix B.3 establishes the existence of an optimal common tax announcement. The implementation probability for the opportunistic type is unique at every $\tau < \tau^O$. At $\tau = \tau^O$, both types implement the same tax rate, and government reputation remains p .

When $m < 1$, implementing the announced tax raises the government's reputation to $q = p/u > p$. The commitment type's continuation revenue is therefore $V_C(q)$, which exceeds the $V_C(p)$ it receives when both types implement the announced tax with certainty. This gain in future revenue gives the commitment type a reason to choose an announced tax that the opportunistic type may revise.

5.3 Government Reputation, Tax Policy, and Investment

After implementation of a first-period tax announcement $\tau < \tau^O$, firms enter the second period with government reputation $q = p/u > 0$. A tax revision reduces government reputation to zero. Proposition 1 gives the following comparisons of second-period investment, expected tax rates, and the expected shares of profits transferred abroad after each policy history,

$$\begin{aligned} K(\tau^R(q), q) &> K_0, \\ q\tau^R(q) + (1 - q)\tau^O &< \tau^O, \\ qy(\tau^R(q)) + (1 - q)y(\tau^O) &< y(\tau^O). \end{aligned} \quad (17)$$

The right-hand sides describe outcomes after a first-period tax revision. The last inequality follows from $\tau^R(q) < \tau^O$ because the share of profits shifted abroad increases

with the tax rate. At the time of the tax revision, raising the tax from τ to τ^O also raises that share at fixed installed capital when $\tau^* \leq \tau < \tau^O$. In the second period, however, lower investment and profits can offset the higher expected share. The comparisons therefore do not determine the change in the total amount shifted.

The expected tax rates and shares of profits transferred abroad reflect firms' uncertainty about government type. Conditional on the government being opportunistic, its terminal tax is τ^O after either first-period policy. When both types implement the first-period announced tax with certainty, the comparison with a tax revision describes the consequences of a deviation, not an observed equilibrium outcome.

Because lower second-period investment reduces the opportunistic type's revenue, a lower first-period tax can be sustained even when initial government reputation is small. With profit shifting, a small tax cut below τ^O raises firms' after-tax returns to first order. The opportunistic type's current gain from reversing that cut is only second order because τ^O maximizes tax revenue on installed capital. The future revenue lost after a tax revision can therefore support a tax reduction that attracts additional investment. I denote the optimal announced tax in (15) by $\tau_1^R(p; \beta)$.

Proposition 3 (Optimal tax announcements at low government reputation). *In the two-period economy with profit shifting, for every fixed $\beta \in (0, 1)$, there is $\bar{p}_\beta > 0$ such that the optimal common tax announcement is unique and strictly decreasing in p on $(0, \bar{p}_\beta)$. As government reputation approaches zero, the optimal announced tax converges to τ^O from below. The opportunistic type's implementation probability has a strictly positive limit, which can be below one, and investment converges to $K_0 > 0$.*

Proof. See Appendix B.3. □

Figure 5 shows how the optimal announced tax and investment losses vary over a wider range of government reputation. I measure losses relative to each economy's full commitment allocation because profit shifting also affects investment under full commitment. With full-commitment investment denoted by K_F , the losses in first-period investment and output in the host country are $1 - K/K_F$ and $1 - (K/K_F)^\alpha$.

With profit shifting, both government types implement the optimal first-period announced tax at every level of government reputation, as shown in Figure 5. The announced tax declines until it reaches the full-commitment rate. At $p = 0.5$, both types implement the announced tax rate in each economy. However, first-period investment is 52.4 percent below its full-commitment level without profit shifting and 3.8 percent below with profit shifting. In each economy, the government announces a rate above the full-commitment rate to reduce the opportunistic type's gain from raising the tax after investment. Firms face this higher rate with certainty and consequently invest less

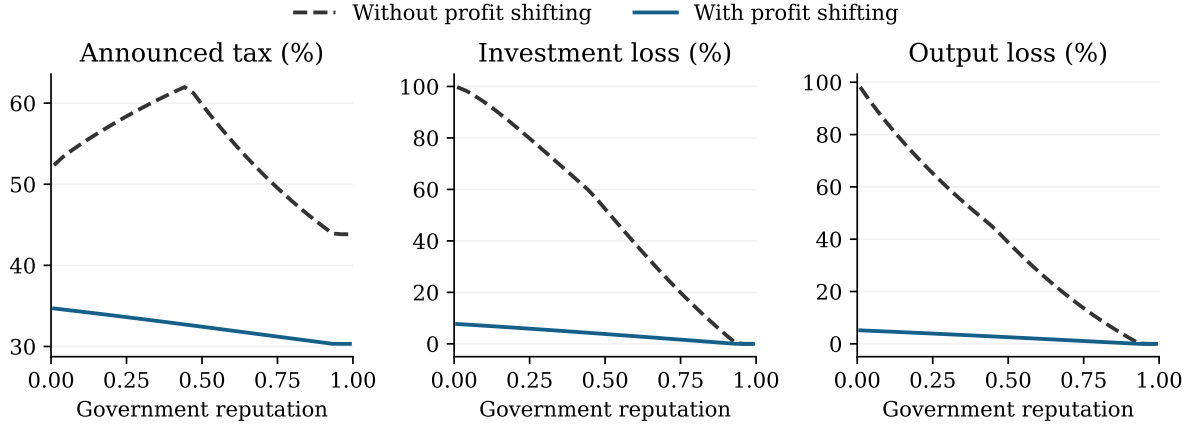


FIGURE 5: Optimal announced taxes and real activity under uncertain commitment

Notes: The common tax announcement maximizes the commitment type's discounted revenue subject to both announcement incentive compatibility constraints and the implementation incentive compatibility conditions. The discount factor is 0.6. Profit-shifting costs are incurred and deducted in the country from which profits are transferred, with $\gamma = 1.25$. Other parameters are those of Section 4. The losses in first-period investment and output in the host country are measured against each economy's full-commitment allocation.

than they would under full commitment. The output comparison concerns production in the host country since capital not invested there remains available abroad.

Without profit shifting, the optimal announced tax first rises and then falls with government reputation. On the rising segment, the opportunistic type sometimes revises the announced tax. The probability u of facing the announced rate increases enough to raise $u(1 - \tau_1^R)$ and investment despite the higher tax. Once the opportunistic type implements the optimal announced tax with certainty, higher government reputation increases the future revenue cost of a tax revision and allows for a lower announced tax.

Profit shifting limits the extra revenue the government can obtain by raising the tax on installed capital. It also allows firms to earn positive returns after the government's reputation falls to zero, so investment and tax revenue remain positive in the following period. The opportunistic type's future revenue loss is therefore the difference between its second-period revenue after implementing the announced tax and what it earns in that period after revealing its type.

Higher profit-shifting costs change firms' returns at a fixed tax and can also change the tax that the opportunistic type is willing to implement. Table 2 shows the consequences of raising these costs before the government announces its tax. At the original announced rate, the current gain from a tax revision rises more than the discounted future revenue loss. The opportunistic type would therefore revise that tax. The government chooses a higher announced rate to reduce the extra revenue available

from a subsequent increase. At $p = 0.1$, investment falls by 5.48 percent after this tax adjustment compared with 0.95 percent if the original tax were implemented.

Under full commitment, the same cost increase reduces investment by 3.67 percent. The larger decline at low government reputation reflects a larger increase in the announced tax. Both government types implement the selected rate before and after the cost change, so uncertainty about commitment affects investment through the chosen tax rather than through more frequent tax revisions. The comparison shows that higher profit-shifting costs can discourage investment through the government's tax adjustment as well as through the direct reduction in firms' returns at a fixed tax. Appendix C.3 decomposes the implementation incentive and examines other parameter values.

TABLE 2: Tax and investment responses to higher profit-shifting costs

Government reputation	Tax change (pp)	Investment change (%)		Current revenue change (%)
		At original tax	After tax adjustment	
0.1	2.39	-0.95	-5.48	3.48
0.5	2.05	-0.72	-4.63	3.87
0.9	1.69	-0.53	-3.75	4.02

Notes: The cost parameter rises from $\gamma = 1.25$ to 1.50 before the first tax announcement and remains higher in both periods. The discount factor is 0.6. Other parameters are those of Section 4. Tax changes are in percentage points (pp), and all changes refer to the first period. Percentage changes are relative to the original investment or revenue level. The original-tax column holds the announced tax at its original level and assumes certain implementation. At that tax, the opportunistic type would prefer a tax revision under the higher cost. After the tax adjustment, both types implement the newly announced tax rate. Current revenue is expected revenue across government types. Under full commitment, the investment changes at the original and newly chosen taxes are -0.52 and -3.67 percent.

6 Sustainable Tax Policy and Investment

Implementing the announced tax does not, by itself, deliver the investment available under full commitment. In the numerical comparisons with profit shifting, both types implement a tax above the full-commitment rate at low government reputation. The higher rate reduces the opportunistic type's gain from a further increase but also lowers firms' returns. I characterize how government reputation affects this tax choice and the resulting investment. I then examine whether future investment can substitute for the protection provided by profit shifting when firms continue investing and learning about government type.

Following [Chari and Kehoe \(1990\)](#), a sustainable plan specifies government policies and private decisions after every public policy history. Firms' investment and reporting choices must maximize expected discounted profits, subject to the capital and tax-base constraints. Each government type must maximize discounted revenue when choosing its tax announcement. The commitment type must implement the announced tax rate, while the opportunistic type must also maximize discounted revenue when choosing the implemented tax. Beliefs must satisfy Bayes' rule at histories reached with positive probability. Off-path tax announcements lead to a government reputation of zero and the continuation equilibrium specified in [Section 5.2](#).

6.1 Sustainable Taxes in Two Periods

I first study sustainable plans in which both types implement the first-period announced tax. The second period follows the selected one-period equilibrium, in which the opportunistic type may revise the announced tax. Sustainability requires the government's planned actions to be optimal at each period and state, including the terminal tax choice.

When both types implement the first-period announced tax, $u = 1$ and $q = p$. At a given tax, current investment is $K(\tau, 1)$ and does not depend on government reputation. Current revenue is $H(\tau) = \mathcal{R}(K(\tau, 1), \tau)$. The current gain from a tax revision, $G(\tau)$, and the future revenue loss, $L(p)$, enter the implementation incentive compatibility constraint

$$\underbrace{\mathcal{R}(K(\tau, 1), \tau^O) - H(\tau)}_{G(\tau)} \leq \beta \underbrace{[V_O(p) - R_0]}_{L(p)}. \quad (18)$$

Both current revenues are evaluated at the capital installed after the tax announcement. Firms can shift reported profits after a tax revision but cannot change that capital. The profit transfers limit the extra revenue from the tax increase. Profit shifting also protects firms' returns after the opportunistic type is revealed, ensuring that investment and revenue remain positive in the next period. The future revenue loss is the decline caused by lower investment once firms are aware that the government will impose the discretionary tax.

The announcement incentive compatibility constraints are

$$\begin{aligned} H(\tau) + \beta V_C(p) &\geq (1 + \beta)R_0, \\ H(\tau) + \beta V_O(p) &\geq (1 + \beta)R_0. \end{aligned} \quad (19)$$

Continuation revenue is independent of the first-period tax when both types implement the announced tax. Given the terminal equilibrium, the commitment type, therefore, chooses

$$\max_{\tau \in [0,1]} H(\tau) \quad \text{subject to } G(\tau) \leq \beta L(p) \text{ and (19).} \quad (20)$$

I denote the optimal sustainable tax by $\tau_1^{R,s}(p)$. It need not equal the optimal announced tax τ_1^R in Section 5.2, which allows the opportunistic type to revise the announced tax with a positive probability.

Proposition 4 (Optimal sustainable taxes with profit shifting). *In the economy with profit shifting, for any $\beta \in (0, 1)$, the optimal sustainable tax $\tau_1^{R,s}(p)$ is unique. It equals the full-commitment tax $\tau^R(1)$ if $\tau^R(1)$ satisfies (18). Otherwise, $\tau_1^{R,s}(p)$ is the unique tax in $(\tau^R(1), \tau^O)$ at which the constraint binds. For every $0 < p < p' < 1$,*

$$\tau_1^{R,s}(p') \leq \tau_1^{R,s}(p), \quad \text{strictly if } \tau_1^{R,s}(p) > \tau^R(1).$$

Both announcement incentive compatibility constraints hold at $\tau_1^{R,s}(p)$.

Proof. See Appendix B.4. □

When the full-commitment rate cannot be sustained, the government announces a higher rate, reducing the extra revenue it could obtain by raising the tax after investment. Higher p raises second-period investment and hence the opportunistic type's revenue $V_O(p)$ after the implementation of the first-period announced tax. Revenue after a tax revision remains R_0 . The larger future loss $L(p)$ permits a lower announced tax while the implementation incentive compatibility constraint binds. This tax reduction raises current investment and revenue. Once the full-commitment rate is sustainable, further increases in p leave current taxes and investment unchanged.

I compare this result with an economy without profit shifting to determine what future revenue alone can sustain. The opportunistic type in that economy can collect all current profits earned in the host country by raising an announced tax below one to a rate of one. The tax revision reveals the government's type and eliminates second-period investment, creating a future revenue cost even without profit shifting.

Proposition 5 (Optimal sustainable taxes without profit shifting). *In the economy without profit shifting, under the production assumptions of Proposition 2, the optimal sustainable tax is unique. It equals $\tau^R(1)$ when the full-commitment rate is sustainable. Otherwise, it is the unique tax above $\tau^R(1)$ satisfying*

$$(1 - \tau_1^{R,s})\rho(K(\tau_1^{R,s}, 1)) = \beta\rho(K(\tau^R(p), p)). \quad (21)$$

As government reputation rises, the optimal sustainable tax weakly decreases and investment weakly increases, with strict changes when the tax exceeds $\tau^R(1)$. Both announcement incentive compatibility constraints hold.

Proof. See Appendix B.5. □

In both economies, higher government reputation raises current investment by allowing a lower sustainable tax while the implementation incentive compatibility constraint binds. At a fixed tax implemented by both types, government reputation leaves current investment unchanged. The investment benefit of higher government reputation in these plans comes from the lower tax it permits. An economy can therefore have no current tax revisions and still attract less investment than under full commitment because the government chooses a high rate to make implementation attractive.

The sustainable tax responds in the same direction in both economies, whereas the static model gives opposite responses to the announced tax with and without profit shifting. With future investment, the government compares the current gain from raising the tax with the revenue it would lose after its type is revealed. Profit shifting affects both amounts. It limits the extra revenue from taxing installed capital and allows investment and tax revenue to remain positive after a tax revision. The numerical cost comparison in Section 5.3 illustrates how changes in these incentives alter the selected tax and the investment loss relative to full commitment.

6.2 Sustainable Plans over Multiple Periods

The two-period economy gives the government one future investment decision to consider when choosing whether to raise the current tax. I extend the model to multiple periods to characterize sustainable tax policies and investment throughout the relationship. A tax revision now affects the revenue from all remaining investment decisions.

The economy now lasts T periods, indexed by $t = 1, \dots, T$, with full depreciation and the same access to capital in each period. I consider plans in which both types implement the commonly announced tax at every date $t < T$. At date T , the government

and firms follow the selected one-period equilibrium. Government reputation remains p as long as both types implement the plan. A tax revision reveals the opportunistic type, after which firms expect discretionary taxation in every subsequent period.

Given the terminal equilibrium, I determine the commitment type's preferred sustainable tax by backward induction from date $T - 1$ to date 1. At each date, the opportunistic type's future revenue determines its incentive to implement the currently announced tax rate.

The opportunistic type's revenue from dates t through T , in excess of what it would earn if its type were known before the date- t investment, is denoted by $E_t(p)$. Both revenues are discounted to date t . The terminal value is $E_T(p) = V_O(p) - R_0$. At each earlier date, the optimal sustainable tax $\tau_t^{R,s}(p)$ and excess revenue satisfy

$$\begin{aligned}\tau_t^{R,s}(p) &= \arg \max_{\tau} H(\tau) \quad \text{subject to } G(\tau) \leq \beta E_{t+1}(p), \\ E_t(p) &= H(\tau_t^{R,s}(p)) - R_0 + \beta E_{t+1}(p), \quad t = T - 1, \dots, 1.\end{aligned}\tag{22}$$

When $T = 2$, $E_2(p) = L(p)$, and the recursion gives the two-period sustainable tax $\tau_1^{R,s}(p)$.

Proposition 6 (Sustainable taxes and investment over a finite horizon). *Under the assumptions of Proposition 4 or 5, for any fixed finite horizon $T \geq 2$, (22) selects a unique tax $\tau_t^{R,s}(p)$ at each date $t < T$. Each tax satisfies both types' announcement incentive compatibility constraints and is weakly decreasing in government reputation, strictly so whenever it exceeds $\tau^R(1)$. As $p \downarrow 0$, the tax at each date $t < T$ converges to τ^O with profit shifting and to one without profit shifting. Investment, production in the host country, and revenue at each such date converge to $K_0 > 0$, $zK_0^\alpha > 0$, and $R_0 > 0$ with profit shifting and to zero without profit shifting.*

Proof. See Appendix B.6. □

Higher government reputation continues to permit lower taxes and greater investment while the implementation incentive compatibility constraint binds. As government reputation approaches zero, the future revenue lost after a tax revision vanishes and sustainable taxes approach the discretionary rate. Firms still invest in the baseline economy because profit shifting protects part of their returns even under that rate. The government can therefore attract investment despite having little future revenue to lose from a tax revision. The result so far concerns plans in which both types implement announced taxes before the final period.

6.3 Investment at Low Government Reputation

When both types implement every announced tax before the final period, firms learn nothing about the government type along the equilibrium path. Other pooling equilibria can allow for occasional tax revisions. In these cases, implementing the announced tax raises government reputation and may encourage future investment. I examine whether this learning can prevent investment from vanishing in the economy without profit shifting as initial government reputation approaches zero. Allowing any common tax announcement establishes whether the investment difference depends on the earlier selection of tax announcements.

I allow the common tax announcement and the implementation probability of the opportunistic type to depend on public policy history. Government and firm decisions must remain sequentially optimal, and beliefs must satisfy Bayes' rule. The set of these equilibria over T periods with initial government reputation p is denoted by $\mathcal{E}_T(p)$, and K_e is date-1 investment in equilibrium e . The set includes all terminal pooling equilibria.

Proposition 7 (Investment as government reputation vanishes). *For any fixed finite horizon $T \geq 1$ and $\beta \in (0, 1)$, under the production assumptions of Proposition 2, investment without profit shifting satisfies*

$$\lim_{p \downarrow 0} \sup_{e \in \mathcal{E}_T(p)} K_e = 0.$$

Date-1 production in the host country and tax revenue consequently converge to zero. With profit shifting as defined in Section 3, there is a constant $\underline{K} > 0$ such that every equilibrium in $\mathcal{E}_T(p)$ satisfies $K_e \geq \underline{K}$, independently of p and T . The comparison does not depend on which common tax announcement the commitment type prefers.

Proof. See Appendix B.7. □

In the baseline economy, firms retain positive returns on investment in the host country, even under a confiscatory tax rate of 100%. Reporting profits abroad therefore sustains investment independently of the government's incentive to implement an announced tax rate.

Removing profit shifting illustrates why this protection matters for a government with a low reputation. To sustain investment that is bounded away from zero, the opportunistic type must leave firms some profits earned in the host country. Implementing a tax below one sacrifices current revenue. As initial government reputation approaches zero, the future revenue preserved by implementing the announced tax rate becomes too small to compensate for that sacrifice. Alternative common tax announcements and history-dependent implementation cannot prevent investment from vanishing in this counterfactual economy.

These results distinguish the investment sustained by firms' ability to protect returns from the additional investment attracted by a lower tax. Profit shifting leaves firms willing to invest even when the government has little future revenue to lose from a tax revision. In sustainable plans, the government may still choose a tax above the rate under full commitment, reducing firms' returns and investment. Higher government reputation permits a lower tax while the implementation incentive compatibility constraint binds and attracts investment beyond the level protected by profit shifting.

7 Concluding Remarks

The optimal corporate tax depends on how firms can protect their returns if the government raises the tax after investment. In the one-period economy, firms earn positive returns under discretionary taxation because they can report profits abroad. Higher government reputation then makes a larger share of expected returns depend on the announced rate. The tax base becomes more elastic with respect to that rate, and the government chooses a lower tax. Without profit shifting, higher government reputation instead makes the tax base less elastic and raises the optimal announced tax under the maintained production assumptions.

Extending the time periods of the model economy, I show that the government also considers the future revenue it would lose by raising the tax after capital is installed. Profit shifting affects both this future loss and the current revenue gain from a tax increase. Among the taxes that both types implement, the government chooses a rate above the full-commitment rate when the future revenue loss is too small to sustain that rate. The higher announced tax reduces its gain from a subsequent increase. Higher government reputation permits a lower tax while this constraint binds. Tax rates and investment can therefore differ based on government reputation, even when both types implement the announced tax. The numerical cost comparison illustrates how changes in firms' ability to shift profits affect this tax choice.

The model with multiple periods shows how repeated investment and profit shifting support investment when government reputation is low. Future tax revenue can make the government willing to implement a low rate, but this incentive weakens as government reputation approaches zero. Without profit shifting, investment then vanishes even when tax policies depend on public history. With profit shifting, firms continue investing because they retain part of their returns after a tax increase. Higher government reputation can still attract additional investment by permitting a lower tax that both types implement.

References

- Abadie, A., Diamond, A., and Hainmueller, J. (2010). Synthetic control methods for comparative case studies: Estimating the effect of California's tobacco control program. *Journal of the American Statistical Association*, 105(490):493–505.
- Aguiar, M. and Amador, M. (2011). Growth in the shadow of expropriation. *The Quarterly Journal of Economics*, 126(2):651–697.
- Altshuler, R., Boller, L., Roberts, K. A., and Suárez Serrato, J. C. (2025). Tax planning and multinational behavior. Working paper, February.
- Anand, R. and Tulin, V. (2014). Disentangling India's investment slowdown. IMF Working Paper 14/47, International Monetary Fund.
- Backus, D. and Driffill, J. (1985). Rational expectations and policy credibility following a change in regime. *The Review of Economic Studies*, 52(2):211–221.
- Beer, S., de Mooij, R., and Liu, L. (2018). International corporate tax avoidance: A review of the channels, effect sizes, and blind spots. Technical Report 18/168, International Monetary Fund.
- Bilicka, K. A., Devereux, M. P., and Güçeri, İ. (2024). Tax policy, investment and profit shifting. Technical Report 33132, National Bureau of Economic Research.
- Chari, V., Kehoe, P., and Prescott, E. (1988). Time consistency and policy. Staff Report 115, Federal Reserve Bank of Minneapolis.
- Chari, V. V. and Kehoe, P. J. (1990). Sustainable plans. *Journal of Political Economy*, 98(4):783–802.
- de Mooij, R. and Liu, L. (2020). At a cost: The real effects of transfer pricing regulations. *IMF Economic Review*, 68:268–306.
- Devereux, M. P., Lockwood, B., and Redoano, M. (2008). Do countries compete over corporate tax rates? *Journal of Public Economics*, 92(5):1210–1235.
- Dovis, A. and Kirpalani, R. (2020). Fiscal rules, bailouts, and reputation in federal governments. *American Economic Review*, 110(3):860–88.
- Dovis, A. and Kirpalani, R. (2021). Rules without commitment: Reputation and incentives. *The Review of Economic Studies*, 88(6):2833–2856.
- Dyrda, S., Hong, G., and Steinberg, J. B. (2024a). A macroeconomic perspective on taxing multinational enterprises. *Journal of International Economics*, 152:104022.
- Dyrda, S., Hong, G., and Steinberg, J. B. (2024b). Optimal taxation of multinational enterprises: A Ramsey approach. *Journal of Monetary Economics*, 141:74–97.

- Forbes, K. J. and Warnock, F. E. (2021). Capital flow waves—or ripples? extreme capital flow movements since the crisis. *Journal of International Money and Finance*, 116:102394.
- Guvenen, F., Mataloni, Jr., R. J., Rassier, D. G., and Ruhl, K. J. (2022). Offshore profit shifting and aggregate measurement: Balance of payments, foreign investment, productivity, and the labor share. *American Economic Review*, 112(6):1848–1884.
- Hong, Q. and Smart, M. (2010). In praise of tax havens: International tax planning and foreign direct investment. *European Economic Review*, 54(1):82–95.
- Hopland, A. O., Lisowsky, P., Mardan, M., and Schindler, D. (2018). Flexibility in income shifting under losses. *The Accounting Review*, 93(3):163–183.
- International Monetary Fund (2007). Haiti: Selected issues and statistical appendix. Country Report 07/292, International Monetary Fund.
- International Monetary Fund (2008). Zambia: Request for a three-year arrangement under the poverty reduction and growth facility. IMF Country Report 08/187, International Monetary Fund.
- International Monetary Fund (2010). Togo: Third review under the three-year arrangement under the poverty reduction and growth facility. Country Report 10/50, International Monetary Fund.
- Janeba, E. (2000). Tax competition when governments lack commitment: Excess capacity as a countervailing threat. *American Economic Review*, 90(5):1508–1519.
- Janeba, E. (2002). Attracting FDI in a politically risky world. *International Economic Review*, 43(4):1127–1155.
- Johannesen, N. (2010). Imperfect tax competition for profits, asymmetric equilibrium and beneficial tax havens. *Journal of International Economics*, 81(2):253–264.
- Jordà, O. (2005). Estimation and inference of impulse responses by local projections. *American Economic Review*, 95(1):161–182.
- Kallen, C. (2023). Profit shifting and multinational investment. Working paper, University of Wisconsin–Madison, January 14.
- King, R. G., Lu, Y. K., and Pastén, E. S. (2008). Managing expectations. *Journal of Money, Credit and Banking*, 40(8):1625–1666.
- Lu, Y. K. (2013). Optimal policy with credibility concerns. *Journal of Economic Theory*, 148(5):2007–2032.
- Mardan, M. and Stimmelmayer, M. (2020). Tax competition between developed, emerging, and developing countries—same same but different? *Journal of Development Economics*, 146:102491.

- McClure, C. G. (2023). How costly is tax avoidance? evidence from structural estimation. *The Accounting Review*, 98(6):353–380.
- Ministry of Finance, Government of India (2012). Memorandum explaining the provisions in the finance bill, 2012. Union Budget 2012–13, March 16, p. 19.
- Ministry of Finance, Government of India (2021). Framing of rules for the amendments made by the taxation laws (amendment) act, 2021. Press Information Bureau, 28 August. <https://www.pib.gov.in/Pressreleaseshare.aspx?PRID=1749947>.
- Mongrain, S., Oh, D., and van Ypersele, T. (2023). Tax competition in the presence of profit shifting. *Journal of Public Economics*, 224:104940.
- OECD (2026). Corporate income tax rates database. Combined statutory corporate income tax rates, accessed September 22, 2026.
- Ortigueira, S. and Pereira, J. (2022). Lack of commitment, retroactive taxation, and macroeconomic instability. *Journal of the European Economic Association*, 20(1):264–311.
- Peralta, S., Wauthy, X., and van Ypersele, T. (2006). Should countries control international profit shifting? *Journal of International Economics*, 68(1):24–37.
- Phelan, C. (2006). Public trust and government betrayal. *Journal of Economic Theory*, 130(1):27–43.
- Phelan, C. and Stacchetti, E. (2001). Sequential equilibria in a ramsey tax model. *Econometrica*, 69(6):1491–1518.
- Schnitzer, M. (1999). Expropriation and control rights: A dynamic model of foreign direct investment. *International Journal of Industrial Organization*, 17(8):1113–1137.
- The PRS Group (n.d.). ICRG methodology. Methodology document, accessed September 22, 2026.
- Thomas, J. and Worrall, T. (1994). Foreign direct investment and the risk of expropriation. *The Review of Economic Studies*, 61(1):81–108.
- Tørsløv, T., Wier, L., and Zucman, G. (2023). The missing profits of nations. *The Review of Economic Studies*, 90(3):1499–1534.
- World Bank (2004). *World Development Report 2005: A Better Investment Climate for Everyone*. World Bank and Oxford University Press, Washington, DC.

A Data Appendix

A.1 Sources and Variable Construction

The risk measure is constructed from the monthly Investment Profile scores in ICRG Table 3B. The annual measure is twelve minus the annual mean of those scores.

Economic controls are real GDP per capita in constant 2015 US dollars, population, and natural-resource rents/GDP from the World Bank. The additional ICRG controls are annual averages lagged one year. I subtract corruption and law and order scores from six, bureaucratic quality from four, and government stability and internal and external conflict from twelve. Six, four, and twelve are the respective component maxima in the ICRG methodology.

A.2 Tax Estimates and Robustness

Table 3 reports all slope coefficients for columns (4) and (5) of Table 1. Table 4 examines the sensitivity of the risk coefficient to including the zero tax rates reported for Haiti and Togo.

Adding lagged GDP growth leaves the investment-risk coefficient positive and significant at the 5 percent level in all columns of Table 4. With economic controls alone, including the zero-rate observations gives a coefficient of 0.263, which is statistically insignificant. Excluding those observations gives 0.395, significant at the 1 percent level.

The estimated risk coefficient remains positive when I exclude each country in turn. For the specification in Table 4, column (2), the coefficients range from 0.371 to 0.457 and remain significant at the 1 percent level. No single country accounts for the positive association.

A.3 Sensitivity of the Investment Regressions

Figure 6 adds the institutional and political controls to the local projections in Figure 1. The following-year coefficient is -0.234 , compared with -0.243 with economic controls alone. The estimated direction and timing are similar across the two specifications.

I also estimate (2) with inflows in levels and with the inverse hyperbolic sine transformation, $\text{asinh}(F)$. The transformation is applied without rescaling monetary flows, so its coefficient is not a percentage change. The following-year estimate is negative in both specifications. The inverse hyperbolic sine estimate is statistically insignificant.

TABLE 3: Tax regressions with full control coefficients

	(4)	(5)
Investment risk	0.426*** (0.143)	0.420*** (0.145)
Log real GDP per capita	1.653 (1.337)	1.708 (1.354)
Log population	4.364** (1.894)	4.390** (1.887)
Resource rents/GDP	0.041 (0.033)	0.043 (0.033)
Corruption risk	-0.417 (0.306)	-0.427 (0.305)
Law and order risk	0.013 (0.471)	0.011 (0.473)
Bureaucratic quality risk	1.419 (1.029)	1.419 (1.028)
Government stability risk	0.102 (0.114)	0.098 (0.114)
Internal conflict risk	-0.336 (0.239)	-0.334 (0.240)
External conflict risk	-0.080 (0.311)	-0.088 (0.310)
Lagged GDP growth	–	-0.016 (0.024)
Country and year effects	Yes	Yes
Observations	1,887	1,887

Notes: Columns (4) and (5) of Table 1, using 1,887 observations in 101 countries over 2003–2021. Annual ICRG averages require twelve monthly observations. All ICRG measures are lagged by one year and expressed as risks. Income and population are logged, and resource rents/GDP and statutory rates are in percentage points. Parentheses contain country-clustered standard errors. ***, **, and * denote significance at the 1, 5, and 10 percent levels. Dashes indicate controls omitted from a specification.

Risk scores and investment inflows can respond to the same economic conditions. Inflows may also adjust before a published risk score changes. The regressions describe the timing of risk and investment inflows without identifying the effect of a tax revision.

A.4 Synthetic Control for the India Tax Event

I construct separate synthetic comparisons for investment risk and FDI inflows because the countries that match India’s risk history need not match its investment history. Quarterly risk is the average of monthly ICRG observations. For inflows, I divide each country’s quarterly series by its own pre-event mean. This normalization puts inflows on a comparable scale without taking logarithms.

TABLE 4: Tax regressions under alternative sample restrictions

	(1)	(2)	(3)
Investment risk	0.353** (0.150)	0.426*** (0.143)	0.430*** (0.143)
Sample	Original	Main	Exclude Haiti/Togo
Full controls	Yes	Yes	Yes
Country and year effects	Yes	Yes	Yes
Observations	1,907	1,887	1,869
Countries	101	101	99
Within R^2	0.058	0.038	0.039

Notes: OECD annual rates over 2003–2021. All columns include economic, institutional, and political controls and country and year effects. Column (1) uses rates as recorded. Column (2) excludes twenty zero-rate observations in Haiti (2003–2010) and Togo (2003–2014). International Monetary Fund reports document positive rates during these intervals, including Haiti’s 30 percent rate after its 2006 reform (International Monetary Fund, 2007) and Togo’s positive rates before and after its 2009–2010 reductions (International Monetary Fund, 2010). Other jurisdictions with zero tax rates remain included. Column (3) excludes Haiti and Togo entirely. I also reestimate columns (1) and (2), excluding each country in turn. Parentheses contain country-clustered standard errors. ***, **, and * denote significance at the 1, 5, and 10 percent levels.

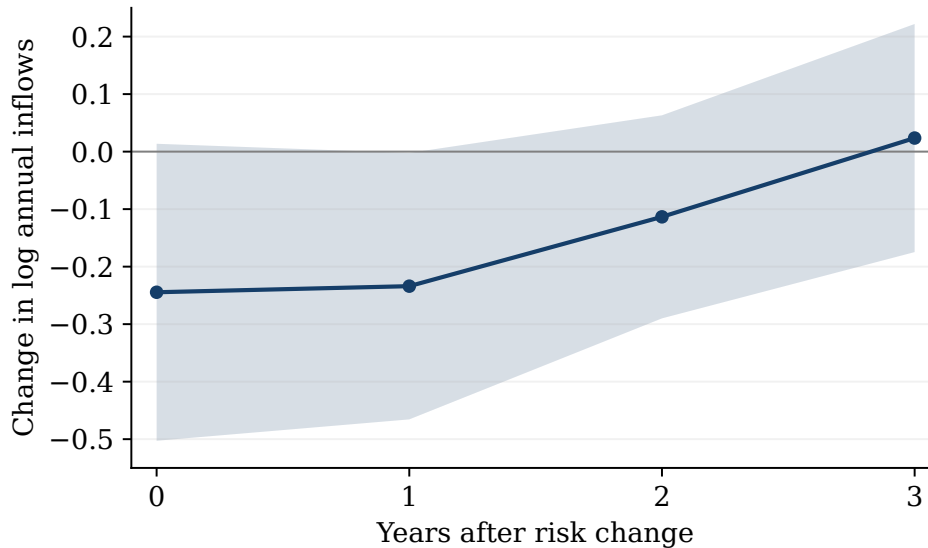


FIGURE 6: FDI Inflows after an Increase in Investment Risk with Additional Controls

Notes: The line plots β_s from (2) with all controls, using 628 observations in 55 countries over the dates specified in Figure 1. The additional controls are lagged ICRG risks for corruption, law and order, bureaucratic quality, government stability, internal conflict, and external conflict. The shaded band is a pointwise 95 percent confidence interval using country-clustered standard errors. Using all observations available for the following-year regression with economic controls gives a coefficient of -0.206 from 790 observations in 56 countries, compared with -0.243 in Figure 1. Levels and inverse hyperbolic sine regressions include nonpositive flows and use 784 observations in 56 countries. The level estimate is significant with country clustering but not with clustering by both country and year.

India's outcome in quarter t is denoted by Y_{It} , and Y_{jt} is the corresponding outcome in donor country j . For inflows, these variables denote the normalized series. The pre-event quarters form the set $\mathcal{T}_0$, and the donor pool is $\mathcal{D}$. For each outcome, the weights solve

$$\begin{aligned}\hat{w} &= \arg \min_{w_j \geq 0, \sum_{j \in \mathcal{D}} w_j = 1} \left\{ \frac{1}{|\mathcal{T}_0|} \sum_{t \in \mathcal{T}_0} \left(Y_{It} - \sum_{j \in \mathcal{D}} w_j Y_{jt} \right)^2 + 10^{-8} \sum_{j \in \mathcal{D}} w_j^2 \right\}, \\ \hat{Y}_{It} &= \sum_{j \in \mathcal{D}} \hat{w}_j Y_{jt}.\end{aligned}\tag{23}$$

The criterion uses the pre-event outcome path alone, with a small quadratic penalty on the weights. The fitted weights remain fixed after the tax announcement. Table 5 reports the complete donor pool and the separate weights for risk and inflows.

TABLE 5: Country Weights in India's Synthetic Comparisons

Country	Weight (%)		Country	Weight (%)		Country	Weight (%)	
	Risk	FDI		Risk	FDI		Risk	FDI
Argentina	2.63	0.00	Guatemala	0.00	0.00	Poland	0.00	0.00
Australia	0.00	0.00	Hong Kong	0.00	0.00	Portugal	1.06	0.00
Austria	0.00	0.00	Hungary	0.00	0.00	Romania	0.00	0.00
Bangladesh	0.00	0.00	Iceland	0.87	0.00	Russia	0.00	0.00
Bolivia	13.24	20.18	Indonesia	0.00	0.00	Singapore	0.00	0.00
Brazil	0.00	0.00	Ireland	0.00	1.01	Slovakia	0.00	0.00
Canada	7.84	0.00	Israel	0.00	0.00	Slovenia	0.00	0.00
Chile	0.00	0.00	Italy	4.43	0.00	South Africa	19.49	13.89
China	0.18	0.00	Japan	0.00	0.00	South Korea	0.00	0.00
Colombia	0.00	0.33	Latvia	0.00	0.00	Spain	0.00	0.00
Costa Rica	6.11	0.00	Lithuania	0.00	0.00	Sri Lanka	0.00	0.00
Croatia	0.00	10.53	Malaysia	0.00	0.00	Sweden	0.00	0.00
Czech Republic	0.00	0.00	Mexico	0.00	0.00	Switzerland	0.00	0.00
Denmark	0.00	0.45	Netherlands	0.00	0.00	Taiwan	0.00	0.00
Estonia	0.00	0.00	New Zealand	0.00	0.86	Thailand	6.90	0.00
Finland	0.00	0.00	Norway	0.00	0.00	Turkey	0.00	0.00
France	0.00	0.00	Panama	0.00	20.22	United Kingdom	0.00	0.00
Germany	0.00	1.16	Peru	0.00	22.45	United States	24.23	0.00
Greece	0.00	8.92	Philippines	0.00	0.00	Venezuela	13.01	0.00

Notes: All 57 donor countries are shown. Eligibility requires complete quarterly risk and inflow observations over 2005Q1–2014Q1 and a positive mean inflow over 2005Q1–2011Q4. Weights are fitted separately for each outcome over these 28 pre-event quarters and sum to 100 percent before rounding. Entries of 0.00 denote weights below 0.005 percent. Inflows are divided by each country's pre-event mean, allowing zero or negative quarterly flows. The transition quarter 2012Q1 is excluded from fitting; the comparison after the tax announcement covers 2012Q2–2014Q1. Quarterly risk requires all three monthly scores.

Table 6 reports the change in the average difference between India and the synthetic comparison before and after the tax announcement. For the placebo comparisons, I apply the same procedure to each comparison country in turn, excluding India from its

comparison pool. The reported ranks compare the magnitude of India's change with those of the donor countries.

TABLE 6: India relative to the synthetic comparison

	Risk points	Normalized inflows
Post-minus-pre mean gap	1.046	-0.732
Pre-event root mean squared gap	0.108	0.233
Post/pre root mean squared gap	9.680	3.350
Rank among placebo comparisons	0.172	0.121
2010–2011 mean prediction error	-0.101	-0.266
Asian comparison, post-minus-pre gap	1.437	-0.753

Notes: The donor pool and dates are specified in Table 5. The rank fraction counts India and placebo countries with an absolute change in the average gap at least as large as India's, divided by 58. Placebo fits exclude India. These are descriptive ranks, not random-assignment p -values. For the pre-event prediction, weights and mean inflows used for normalization are estimated over 2005–2009. The mean prediction error is India's outcome minus the synthetic comparison over 2010–2011. The Asian donor pool is Bangladesh, China, Indonesia, Korea, Sri Lanka, Malaysia, the Philippines, and Thailand. Additional checks remove each donor with a baseline weight above one percent in turn.

The synthetic comparison overpredicts India's inflows in the pre-event prediction exercise. Part of the divergence therefore precedes the tax event. The differences after the tax announcement retain their signs when I restrict the donor pool to Asian countries or exclude individual donors. These comparisons document higher risk and lower relative inflows around the tax announcement, with the concurrent policy changes described in Section 2.

B Proofs

B.1 Proof of Proposition 1

Proof. I first bound how the elasticity of profits earned in the host country changes with expected returns. The bound will determine the sign of the cross-partial of log revenue with respect to the announced tax and government reputation. Without inward profit shifting, the share of profits earned abroad that remains after tax is $B = 1 - \tau^*$, and $\mu(k) = -\rho''(k)/\rho'(k)$ measures the proportional decline in marginal profits as capital increases. For a given after-tax return $w > 0$ per unit of profit earned in the host country, investment $\mathcal{K}(w)$ solves

$$w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K}).$$

The production restriction in (6) makes marginal profits positive and strictly decreasing. Their limit at zero is infinite, so $\mathcal{K}(w)$ is unique, interior, and strictly increasing. The

elasticity of profits earned in the host country and the rate at which that elasticity declines are

$$\varepsilon(w) = \frac{d \log \rho(\mathcal{K}(w))}{d \log w}, \quad \chi(w) = -\frac{d \log \varepsilon(w)}{d \log w}. \quad (24)$$

Writing $K = \mathcal{K}(w)$, $\mu_H = \mu(K)$, and $\mu_F = \mu(\bar{k} - K)$, implicit differentiation gives

$$\varepsilon(w) = \frac{\rho'(K)}{\rho(K)(\mu_H + \mu_F)} > 0. \quad (25)$$

Its reciprocal is

$$-\frac{\rho(K)\rho''(K)}{\rho'(K)^2} + \frac{\rho(K)}{\rho'(K)}\mu'(\bar{k} - K).$$

The first term increases with K when $r^* > 0$ and is constant when $r^* = 0$. The ratio $\rho(K)/\rho'(K)$ increases by concavity. Moreover,

$$-\frac{\mu'(k)}{\mu(k)^2} = \frac{1 - (2 - \alpha)r^*/(\alpha z k^{\alpha-1})}{1 - \alpha} > 0, \quad (26)$$

so $\mu(\bar{k} - K)$ also increases with K . Thus ε decreases strictly with w and $\chi > 0$. The reciprocal of ε tends to $(1 - \alpha)/\alpha$ as $K \downarrow 0$.

The bound below uses the relative curvature $\lambda = \mu_H/(\mu_H + \mu_F)$. Log differentiation of (25) gives

$$\chi(w) = \lambda \left[1 + \frac{\rho'(K)}{\rho(K)\mu_H} \right] + \frac{\mu'(K) - \mu'(\bar{k} - K)}{(\mu_H + \mu_F)^2}.$$

Positive marginal profits imply

$$\frac{\rho'(K)}{\rho(K)\mu_H} = \frac{\alpha}{1 - \alpha} \frac{[1 - r^*/(\alpha z K^{\alpha-1})]^2}{1 - r^*/(z K^{\alpha-1})} \leq \frac{\alpha}{1 - \alpha}.$$

Using (26) for both countries therefore yields

$$\chi(w) \leq \frac{1}{1 - \alpha} \left[1 - \lambda + (2 - \alpha)\lambda^2 \frac{r^*}{\alpha z K^{\alpha-1}} \right].$$

When $K \leq \bar{k}/2$, $\mu_H \geq \mu_F$, so $1 - \lambda \leq 1/2$ and

$$\chi(w) \leq C_\rho \equiv \frac{1}{1 - \alpha} \left[\frac{1}{2} + \frac{(2 - \alpha)r^*}{\alpha z (\bar{k}/2)^{\alpha-1}} \right]. \quad (27)$$

Since $\varepsilon(w) = \lambda \rho'(K)/[\rho(K)\mu_H]$, the preceding inequalities also give $\varepsilon(w) \leq \alpha/(1 - \alpha)$ at every investment allocation.

Under outward profit shifting, the share of profits earned in the host country that remains taxable there after profit transfers and cost deductions is $b(\tau) = 1 - y(\tau) -$

$\gamma y(\tau)^2/2$, so $v(\tau) = \tau b(\tau)$ and $A'(\tau) = -b(\tau)$. Before the tax base in the host country vanishes,

$$v''(\tau) = -\frac{B^2(\tau + 2)}{\gamma(1 - \tau)^4} < 0. \quad (28)$$

The second restriction in (6) implies $v'(\tau^*) = 1 - \tau^*/(\gamma B) > 0$. At the tax rate where the tax base in the host country first reaches zero, $b = 0$ and the left derivative of v is negative. Among tax rates that induce outward profit shifting, government revenue is therefore uniquely maximized at $\tau^O > \tau^*$, whose first-order condition is (11). The after-tax return per unit of profit earned in the host country at the discretionary tax is $A_O = A(\tau^O)$. By transferring a share $\bar{y}$ of profits earned in the host country, the firm retains $B\bar{y}$ per unit of those profits at every tax that induces outward profit shifting. The optimal transfer at τ^O is interior, so

$$A_O > B\bar{y}, \quad A(\tau^*) = B, \quad A(\tau) \geq B\bar{y} \quad \text{for } \tau \geq \tau^*. \quad (29)$$

For $\tau < \tau^*$, the share of profits earned abroad transferred into the host country is $x(\tau) = (\tau^* - \tau)/(B\gamma)$. The second restriction in (6) ensures $x(\tau) < \bar{y}$. The after-tax returns per unit of profit earned in the host country and abroad are $1 - \tau$ and

$$F(\tau) = B + (\tau^* - \tau)x(\tau) - \frac{B\gamma x(\tau)^2}{2}.$$

Since $(1 - \tau) - F(\tau) = (\tau^* - \tau)[1 - x(\tau)] + B\gamma x(\tau)^2/2 \geq 0$, the after-tax return per unit of profit earned in the host country is at least as large as that earned abroad for inward transfers. Together with (29), this implies that the ratio of after-tax returns per unit of profit earned in the host country and abroad is at least $\bar{y}$ at every implemented tax. Taking expectations preserves the inequality. The investment condition therefore gives, for $K \leq \bar{k}/2$,

$$\bar{y} \leq \frac{\rho'(\bar{k} - K)}{\rho'(K)} \leq \left(\frac{K}{\bar{k} - K} \right)^{1-\alpha}, \quad \frac{\rho(K)}{\rho(\bar{k} - K)} \geq \left(\frac{K}{\bar{k} - K} \right)^\alpha.$$

Subtracting the common rental cost weakly lowers the ratio of marginal profits abroad to those in the host country and weakly raises the ratio of profits earned in the host country to those earned abroad, giving the inequalities on the right. Hence

$$\frac{\rho(K)}{\rho(\bar{k} - K)} \geq \bar{y}^{\alpha/(1-\alpha)}.$$

The same bound holds when $K > \bar{k}/2$, because the profit ratio then exceeds one. At every induced capital allocation, revenue at a tax inducing inward profit shifting satisfies

$$\frac{\partial \mathcal{R}(K, \tau)}{\partial \tau} = \rho(K) + \left[x(\tau) - \frac{\tau}{B\gamma} \right] \rho(\bar{k} - K) > 0,$$

where the last inequality follows from the third restriction in (6). Thus the same τ^O is the global discretionary optimum at every capital allocation induced by a tax announcement.

For a tax announcement $\tau < \tau^*$, the expected after-tax returns per unit of profit earned in the host country and abroad are

$$w_H = p(1 - \tau) + (1 - p)A_O, \quad w_F = pF(\tau) + (1 - p)B.$$

Investment is $\mathcal{K}(Bw_H/w_F)$. Since $F'(\tau) = -x(\tau)$,

$$-\frac{1}{1 - \tau} \leq \frac{\partial}{\partial \tau} \log \frac{w_H}{w_F} = -\frac{p}{w_H} + \frac{px(\tau)}{w_F} \leq 0.$$

The upper bound follows from $x(\tau)(1 - \tau) \leq F(\tau)$ and $x(\tau)A_O \leq B$. Using the bounds on the profit ratio and ε gives

$$\frac{1}{\rho(K)} \frac{d}{d\tau} \{ \tau [\rho(K) + x(\tau)\rho(\bar{k} - K)] \} \geq 1 - \frac{\tau^* \alpha}{(1 - \alpha)B} - \frac{\tau^*}{B\gamma} \bar{y}^{-\alpha/(1-\alpha)} > 0.$$

At τ^* , the right-hand derivatives are $b' = -1/(B\gamma)$ and $A' = -1$, so the same positive lower bound applies to the right-hand derivative of revenue. Taxes above τ^O yield lower values of both A and v than at τ^O and hence less revenue. At τ^O , $v' = 0$ and investment decreases with the announced tax for $p > 0$, so the revenue derivative is negative. Every optimal tax announcement is therefore interior to (τ^*, τ^O) .

For a tax in this interval, the expected after-tax return per unit of profit earned in the host country is $w = pA(\tau) + (1 - p)A_O$, and log revenue is

$$f(\tau, p) = \log \{ v(\tau) \rho(\mathcal{K}(w)) \}.$$

Here $w \leq B$, so $\mathcal{K}(w) \leq \bar{k}/2$ and (27) applies. Differentiation gives

$$\begin{aligned} f_\tau &= \frac{v'}{v} + A' \frac{p\varepsilon(w)}{w}, \\ f_{\tau p} &= \frac{A'\varepsilon(w)}{w^2} \{ A_O - p\chi(w)[A(\tau) - A_O] \}. \end{aligned} \tag{30}$$

The fourth restriction in (6) is equivalent to $C_\rho(1/\bar{y} - 1) < 1$. Since $A(\tau) \leq B$ and $p \leq 1$,

$$A_O - p\chi(w)[A(\tau) - A_O] \geq A_O - C_\rho(B - A_O) > B[(1 + C_\rho)\bar{y} - C_\rho] > 0.$$

Thus $f_{\tau p} < 0$.

Uniqueness follows from the curvature of log revenue. The required identities are $v = -\tau A'$ and

$$A''(\tau) = \frac{B^2}{\gamma(1-\tau)^3} > 0, \quad A'''(\tau) = \frac{3B^2}{\gamma(1-\tau)^4} > 0.$$

At a stationary point, $v'/v = -pA'\varepsilon/w$, so the second derivative is

$$f_{\tau\tau} = \frac{A'' + \tau A''' - \tau(A'')^2/A'}{\tau A'} - \left(\frac{v'}{v}\right)^2 \left[1 + \frac{1 + \chi(w)}{\varepsilon(w)}\right] < 0. \quad (31)$$

Every stationary point is a strict maximum. Two maxima would require a stationary minimum between them. The derivative signs at τ^* and τ^O therefore establish a unique global optimum. The implicit-function theorem and (30) give

$$\frac{d\tau^R(p)}{dp} = -\frac{f_{\tau p}}{f_{\tau\tau}} < 0.$$

Continuity extends the strict ordering of optimal announced tax rates to $p' = 1$.

The tax-base elasticity satisfies

$$-\frac{\partial}{\partial \log \tau} \log \left[\frac{\mathcal{R}(K(\tau, p), \tau)}{\tau} \right] = 1 - \tau f_\tau(\tau, p).$$

Its derivative with respect to government reputation is $-\tau f_{\tau p} > 0$ at each fixed tax in (τ^*, τ^O) . At the optimum, $f_\tau = 0$, so the tax-base elasticity equals one.

Both a higher prior and a lower announced tax increase w and investment. At each fixed tax in (τ^*, τ^O) , higher government reputation also raises the commitment type's revenue, so maximized revenue increases strictly. At $p = 0$, $K_0 = \mathcal{K}(A_O)$ and $R_0 = v(\tau^O)\rho(K_0)$. The commitment type can obtain R_0 by announcing τ^O and strictly improves on it at its interior optimum. The opportunistic type earns $v(\tau^O)$ per unit of profit earned in the host country. Investment exceeds K_0 and increases with government reputation, so its revenue exceeds R_0 and rises strictly. Both announcement incentive compatibility constraints hold. Finally, the announced tax and the share of profits transferred abroad under that tax are below their discretionary counterparts

and decline with government reputation. The expected tax rate and expected share of profits transferred abroad consequently decline as well. $\square$

The proof also establishes the equilibrium when firms know that the government is opportunistic. At $p = 0$, the unique investment K_0 and the discretionary tax τ^O are mutual best responses.

For $p > 0$, after the selected common tax announcement, firms retain prior p and choose $K(\tau^R(p), p)$. The commitment type implements the announced tax rate, and the opportunistic type imposes τ^O . After an off-path tax announcement, firms assign probability one to the opportunistic type and invest K_0 . Either type can then collect at most R_0 . The announcement incentive compatibility constraints verified above rule out profitable deviations. Firms optimize reporting after every history, so these strategies and beliefs form a weak perfect Bayesian equilibrium.

I first examine separating tax announcements, which reveal government type before investment, to justify the selection. An opportunistic type identified before investment earns R_0 . If the commitment type announces $\tau < \tau^O$, firms invest more than K_0 , and the opportunistic type gains by imitating that tax announcement before imposing τ^O . If the commitment type announces $\tau > \tau^O$, it earns less than R_0 and can gain by announcing τ^O . Any remaining candidate for a separating equilibrium therefore gives both types R_0 . Both strictly prefer the selected common tax announcement.

Under the strongly coherent criterion of [Lu \(2013\)](#), an off-path tax announcement belonging to an alternative pooling equilibrium that strictly benefits the commitment type must induce a posterior at least as high as p . The commitment type chooses that tax announcement with probability one, while the opportunistic type does so with probability at most one. At any common tax announcement that does not maximize the commitment type's revenue, moving to its maximizing tax announcement gives a strict gain. Revenue at that tax announcement increases with the posterior, so the deviation remains profitable under the required belief. A candidate for a separating equilibrium is ruled out in the same way by a common tax announcement near the optimum and outside the separating path, which benefits both types. Conversely, no alternative pure tax-announcement equilibrium improves the commitment type's payoff at the selected optimum. Only the opportunistic type can prefer an alternative equilibrium, so the off-path belief assigning probability zero to the commitment type satisfies the criterion and supports the selected allocation.

B.2 Proof of Proposition 2

Proof. The investment function and notation are defined in Appendix B.1. For $p > 0$ and $\tau < 1$, investment without profit shifting satisfies (12), so $K = \mathcal{K}(p(1 - \tau))$, $K_\tau < 0$, and $K_p > 0$. The tax-base elasticity is

$$-\frac{\partial \log \rho(K)}{\partial \log \tau} = \frac{\tau}{1 - \tau} \varepsilon(p(1 - \tau)).$$

It decreases with government reputation because ε decreases strictly. At an interior optimum, the elasticity equals one. Equivalently,

$$\tau^R(p) = 1 - \left\{ 1 + \left[-\frac{\rho''(K)}{\rho'(K)} - \frac{\rho''(\bar{k} - K)}{\rho'(\bar{k} - K)} \right] \frac{\rho(K)}{\rho'(K)} \right\}^{-1}. \quad (32)$$

The expression in braces increases strictly with K , as shown in the proof of Proposition 1. Thus the tax rate satisfying the first-order condition increases with capital. Substituting (32) into

$$p = \frac{B}{1 - \tau^R} \frac{\rho'(\bar{k} - K)}{\rho'(K)}$$

gives a strictly increasing function of K . This function tends to zero as $K \downarrow 0$ and to infinity as $K \uparrow \bar{k}$. For every $p > 0$, there is therefore exactly one tax rate satisfying the first-order condition, and both the tax and capital increase with p . Revenue is positive for tax rates strictly between zero and one and is zero at either endpoint, so this rate is the global maximizer. Higher capital also requires a larger expected share of profits earned in the host country retained after tax, $p(1 - \tau^R)$. The expected tax rate $1 - p(1 - \tau^R)$ consequently decreases with government reputation.

A known opportunistic type imposes a tax of one, firms invest no capital in the host country, and revenue is zero. Each type earns positive revenue at the selected tax announcement, so both announcement incentive compatibility constraints hold. The equilibrium and selection argument for Proposition 1 applies with $\tau^O = 1$ and $K_0 = R_0 = 0$. In particular, a tax announcement below one that identifies the commitment type can be profitably imitated by the opportunistic type, while a tax announcement of one yields zero revenue. At any nonmaximizing common tax announcement, the commitment type gains by moving to its maximizing rate, and revenue at that rate increases with the posterior. The same off-path belief therefore supports the selected pooling equilibrium and satisfies Lu's criterion. $\square$

B.3 Proof of Proposition 3

Proof. I first establish the existence of an optimal common tax announcement at any positive prior and then characterize it as the prior approaches zero. The existence argument applies to any $p \in (0, 1)$ and $\beta \in (0, 1)$, with the terminal equilibrium and off-path beliefs in Section 5.2. For $\tau < \tau^O$, investment increases with the implementation probability m . Under outward profit shifting, the current gain from a tax revision is $[v(\tau^O) - v(\tau)]\rho(K(\tau, u))$, which increases with investment. Under inward profit shifting, the transfer per unit of profit earned abroad is denoted by $x(\tau)$. The gain is

$$[v(\tau^O) - \tau]\rho(K(\tau, u)) - \tau x(\tau)\rho(\bar{k} - K(\tau, u)).$$

This also increases with investment because $v(\tau^O) > \tau^* > \tau$ and profits earned abroad fall as capital in the host country rises. Without profit shifting, the gain is $(1 - \tau)\rho(K(\tau, u))$ and has the same property. Meanwhile, $q = p/[p + (1 - p)m]$ falls with m , reducing $V_O(q)$. Hence $D(\tau, m; p)$ is strictly increasing in m for $\tau < \tau^O$.

The implementation probability is therefore uniquely determined by the endpoint inequalities or the interior equation $D = 0$. It is continuous in τ . At $\tau = \tau^O$, both types implement the same tax and government reputation remains p . Since $V_O(p) > R_0$, both types also implement announced taxes sufficiently close to τ^O from below. The allocation is thus continuous on $[0, \tau^O]$. The rate τ^O satisfies both announcement incentive compatibility constraints, so their feasible set is nonempty and closed.

Tax announcements above τ^O are dominated. At such a tax announcement, investment is at most K_0 , current revenue from implementing the announced tax rate is below R_0 , and revenue at the discretionary tax is at most R_0 . When $m = 1$, announcing τ^O gives higher current revenue and the same continuation revenue.

If $m < 1$, the current gain from a tax revision is at least $\beta[V_O(q) - R_0]$, with equality when $m > 0$. Holding u fixed, the gain from a tax revision falls continuously from $v(\tau^O)\rho(K(0, u))$ to zero as the announced tax rises from zero to τ^O . The gain at zero is at least the gain at the original tax announcement. There is therefore a $\tilde{\tau} < \tau^O$ at which the gain equals $\beta[V_O(q) - R_0]$. The same m and q satisfy the implementation incentive compatibility conditions at $\tilde{\tau}$. Investment is now above K_0 , and the commitment type receives the revenue obtainable at the discretionary tax on that larger capital stock, less $\beta[V_O(q) - R_0]$. Its current revenue strictly increases, its continuation revenue is unchanged, and the opportunistic type's payoff weakly increases. Both announcement incentive compatibility constraints remain satisfied. Continuity on the compact feasible set in $[0, \tau^O]$ then establishes existence of an optimal common tax announcement.

The result at low government reputation concerns the economy with profit shifting. As in Appendix B.1, $B = 1 - \tau^*$ and investment $\mathcal{K}(w)$ solves $w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K})$.

Profits earned in the host country are $S(w) = \rho(\mathcal{K}(w))$, with $A_O = A(\tau^O)$ and $K_0 = \mathcal{K}(A_O)$. Equation (28) gives $v''(\tau^O) < 0$. Since $A'(\tau^O) < 0$ and $S'(A_O) > 0$, the local tax response is measured by the positive coefficient

$$\kappa_R = \frac{v(\tau^O)A'(\tau^O)S'(A_O)}{v''(\tau^O)\rho(K_0)} > 0.$$

The terminal tax announcement satisfies

$$v'(\tau^R)S(w) + v(\tau^R)S'(w)qA'(\tau^R) = 0, \quad w = A_O + q[A(\tau^R) - A_O].$$

At $(\tau^R, q) = (\tau^O, 0)$, the left-hand side has derivative $v''(\tau^O)\rho(K_0) \neq 0$ with respect to τ^R . The implicit-function theorem gives $\tau^R(q) = \tau^O - \kappa_R q + O(q^2)$. Thus, for $L(q) = V_O(q) - R_0$ and $J(q) = V_O(q) - V_C(q)$,

$$L(q) = \ell q^2 + O(q^3), \quad J(q) = \frac{1}{2}\ell q^2 + O(q^3), \quad \ell = [-v''(\tau^O)]\rho(K_0)\kappa_R^2 > 0. \quad (33)$$

Indeed, $w - A_O = [-A'(\tau^O)]\kappa_R q^2 + O(q^3)$ and $v(\tau^O) - v(\tau^R) = [-v''(\tau^O)]\kappa_R^2 q^2/2 + O(q^3)$. Proposition 1 also gives $J(q) > 0$ for every $q > 0$, since $\tau^R(q) < \tau^O$.

If an optimizing sequence as $p \downarrow 0$ had $q \geq \delta > 0$, then $u = p/q \rightarrow 0$. The expected after-tax returns per unit of profit earned in the host country and abroad converge uniformly to A_O and B , so investment converges to K_0 , including for tax announcements below τ^* . When $0 < m < 1$, the binding implementation incentive compatibility constraint gives the commitment type the value

$$v(\tau^O)\rho(K) + \beta R_0 - \beta J(q). \quad (34)$$

With $m = 0$, its value is at most this amount with $q = 1$. Continuity and positivity of J on $[\delta, 1]$ therefore put the limiting payoff strictly below $(1 + \beta)R_0$. Announcing τ^O instead yields $R_0 + \beta V_C(p) \rightarrow (1 + \beta)R_0$. When $m = 1$, $q = p \rightarrow 0$ directly. Every optimizing sequence must therefore have $q \rightarrow 0$ and, for sufficiently small p , $m > 0$.

The reporting formulas imply a positive lower bound on investment at every announced tax. A tax bounded away from τ^O would consequently leave a positive gain from a tax revision while $L(q) \rightarrow 0$, violating the implementation incentive compatibility constraint when $m > 0$. Hence every optimal tax announcement approaches τ^O . By the preceding dominance argument, it induces outward profit shifting at sufficiently small p .

Using q as the argument, with $u = p/q$ and $q \geq p$, the binding implementation incentive compatibility constraint is

$$[v(\tau^O) - v(\tau)]S \left(A_O + \frac{p}{q}[A(\tau) - A_O] \right) = \beta L(q). \quad (35)$$

After substituting $\tau = \tau^O - qZ$ and dividing by q^2 , the equation extends smoothly to the origin, where it becomes

$$\frac{[-v''(\tau^O)]\rho(K_0)Z^2}{2} = \beta\ell.$$

Its derivative with respect to Z is nonzero at the positive solution. The implicit-function theorem gives a smooth $Z(p, q)$ with $Z(0, 0) = \kappa_R \sqrt{2\beta}$. At $q = p$, both types implement $\tau = \tau^O - pZ(p, p)$, and the implementation incentive compatibility constraint binds. For sufficiently small p , this tax exceeds $\tau^R(1)$. Current revenue decreases with the tax above $\tau^R(1)$, as established in Appendix B.1, so this rate maximizes the commitment type's revenue among tax announcements implemented by both types.

Along the binding constraint, the commitment type's value is

$$U(p, q) = v(\tau^O)S \left(A_O + \frac{p}{q}[A(\tau^O - qZ(p, q)) - A_O] \right) - \beta J(q) + \beta R_0.$$

Its smooth extension satisfies

$$\begin{aligned} U(0, 0) &= (1 + \beta)R_0, & U_q(0, 0) &= 0, \\ U_p(0, 0) &= v(\tau^O)S'(A_O)[-A'(\tau^O)]\kappa_R \sqrt{2\beta} > 0, \\ U_{qq}(0, 0) &= -\beta\ell < 0. \end{aligned}$$

Thus U is strictly concave in q near the origin. Expanding its first-order condition gives

$$U_q(p, q) = U_{pq}(0, 0)p - \beta\ell q + O((|p| + |q|)^2).$$

The implicit-function theorem gives a unique unconstrained maximizer

$$\hat{q}(p) = \lambda_\beta p + O(p^2), \quad \lambda_\beta = \frac{U_{pq}(0, 0)}{\beta\ell}.$$

The constrained optimum is $q(p) = \max\{p, \hat{q}(p)\}$. The preceding convergence argument rules out other optimal tax announcements at sufficiently small p .

At this optimum, the expected after-tax return per unit of profit earned in the host country satisfies

$$w - A_O = [-A'(\tau^O)]\kappa_R\sqrt{2\beta}p + O(p^2), \quad J(q) = O(p^2).$$

The opportunistic type's value is $v(\tau^O)S(w) + \beta R_0$, and the commitment type's value is lower by $\beta J(q)$. Both exceed $(1 + \beta)R_0$ by

$$v(\tau^O)S'(A_O)[-A'(\tau^O)]\kappa_R\sqrt{2\beta}p + O(p^2) > 0$$

for sufficiently small p . Both announcement incentive compatibility constraints therefore hold.

Substituting $q(p)$ into the tax and implementation probability yields

$$\begin{aligned} \tau_1^R(p; \beta) &= \tau^O - c_\beta p + O(p^2), \\ c_\beta &= \kappa_R\sqrt{2\beta}\max\{1, \lambda_\beta\} > 0, \quad \lim_{p \downarrow 0} m(p) = \frac{1}{\max\{1, \lambda_\beta\}} > 0. \end{aligned} \quad (36)$$

For either $q(p) = p$ or $q(p) = \hat{q}(p)$, the optimal announced tax has derivative converging to $-c_\beta$. If $\lambda_\beta = 1$, both cases have the same strictly negative limit. The optimal announced tax is therefore strictly decreasing at sufficiently small p and approaches τ^O from below. Strict concavity in q and uniqueness of the commitment type's preferred tax among tax announcements implemented by both types imply that the optimal tax announcement is unique. Since $w \rightarrow A_O$, investment converges to $K_0 > 0$. $\square$

B.4 Proof of Proposition 4

Proof. The full-commitment tax is denoted by $\tau^F = \tau^R(1)$. Proposition 1 and its proof establish a unique τ^O , a unique $\tau^F \in (\tau^*, \tau^O)$, and continuation revenues that increase strictly with government reputation. When both types implement a tax in $[\tau^*, \tau^O]$,

$$H(\tau) = v(\tau)\rho(K(\tau, 1)), \quad G(\tau) = [v(\tau^O) - v(\tau)]\rho(K(\tau, 1)).$$

Investment decreases with the tax. Appendix B.1 shows that H increases up to τ^F and decreases thereafter. Since $v' > 0$ on $[\tau^*, \tau^O]$,

$$G'(\tau) = -v'(\tau)\rho(K(\tau, 1)) + [v(\tau^O) - v(\tau)]\rho'(K(\tau, 1))K_\tau(\tau, 1) < 0.$$

Below τ^* , investment decreases and current revenue increases with the tax by the inward-transfer calculation in Appendix B.1. Revenue at the discretionary tax therefore decreases, so G decreases there as well. Thus G decreases continuously to zero at τ^O .

Above τ^O , current revenue is below R_0 because both investment and revenue per unit of profit earned in the host country are lower.

At any $p \in (0, 1)$, both continuation revenues are at least R_0 , and $L(p) > 0$. Any current revenue of at least R_0 therefore satisfies both announcement incentive compatibility constraints. The tax τ^O is feasible and yields R_0 , whereas τ^F yields strictly more. If $G(\tau^F) \leq \beta L(p)$, the unconstrained maximizer τ^F is feasible and uniquely optimal. Otherwise, strict monotonicity of G gives a unique $\tau \in (\tau^F, \tau^O)$ satisfying $G(\tau) = \beta L(p)$. Every lower tax violates the implementation incentive compatibility constraint. Every higher tax up to τ^O yields less revenue, and taxes above τ^O yield less than R_0 . This tax rate is therefore uniquely optimal and satisfies both announcement incentive compatibility constraints.

Higher government reputation raises $L(p)$ and lowers the optimal sustainable tax while the implementation incentive compatibility constraint binds, increasing investment and current revenue. Once τ^F is feasible, further increases in government reputation leave current taxes and investment unchanged. $\square$

B.5 Proof of Proposition 5

Proof. Without profit shifting, $\tau^O = 1$, $R_0 = 0$, and

$$H(\tau) = \tau \rho(K(\tau, 1)), \quad G(\tau) = (1 - \tau) \rho(K(\tau, 1)).$$

Investment is positive and strictly decreasing for $\tau < 1$. Proposition 2 and its proof give a unique revenue maximum at $\tau^F = \tau^R(1)$, with H increasing below τ^F and decreasing above it. Both positive factors in G decrease with the tax, so G is strictly decreasing and converges to zero as $\tau \uparrow 1$.

The opportunistic type loses future revenue $L(p) = \rho(K(\tau^R(p), p))$ after a tax revision. This loss is positive and strictly increasing by Proposition 2. The argument for Proposition 4 applies. The optimum is τ^F when $G(\tau^F) \leq \beta L(p)$ and otherwise the unique tax above τ^F satisfying $G(\tau) = \beta L(p)$. While this constraint binds, the optimal sustainable tax decreases and investment increases with government reputation. Both types earn positive revenue, while an off-path tax announcement yields zero, so both announcement incentive compatibility constraints hold. $\square$

B.6 Proof of Proposition 6

Proof. Along the prescribed path, both types receive the same revenue before the terminal date. Their terminal payoff difference is fixed by the selected one-period equilibrium. A plan for the remaining periods that maximizes the commitment type's

revenue therefore also maximizes the opportunistic type's revenue within this class. The opportunistic type's continuation revenue in excess of what it earns when its type is known is denoted by $E \geq 0$. The preceding two proofs give a unique optimal sustainable tax $\tau(E)$. It equals $\tau^F = \tau^R(1)$ if $G(\tau^F) \leq \beta E$ and otherwise solves $G(\tau(E)) = \beta E$ on $(\tau^F, \tau^O]$. Thus $\tau(E)$ is continuous and nonincreasing, strictly decreasing when it exceeds τ^F . Current revenue $H(\tau(E))$ is continuous, nondecreasing in E , and at least R_0 .

The map $E \mapsto H(\tau(E)) - R_0 + \beta E$ is continuous, strictly increasing, and zero at $E = 0$. The unique static optimum and the maximum theorem imply continuity of the terminal value $E_T(p) = V_O(p) - R_0$. The investment condition gives $E_T(p) \rightarrow 0$ as $p \downarrow 0$, and Propositions 1 and 2 give strict monotonicity. Applying (22) backward from date T shows that each $E_t(p)$ is continuous, strictly increasing, and converges to zero as $p \downarrow 0$. At every date $t < T$, $\tau_t^{R,s}(p) = \tau(E_{t+1}(p))$ therefore decreases with government reputation whenever the implementation incentive compatibility constraint binds above τ^F and converges to τ^O as $p \downarrow 0$. Investment converges to $K_0 > 0$ with profit shifting and to zero without profit shifting, where $\tau^O = 1$. Continuity of production and current revenue gives the remaining limits.

Following an off-path tax announcement at date t , firms invest at government reputation zero, and either type can earn at most $R_0 \sum_{j=0}^{T-t} \beta^j$. Revenue at each prescribed tax before date T and both terminal revenues are at least R_0 , so both types' announcement incentive compatibility constraints hold at every date. The recursion imposes the implementation incentive compatibility constraint. After a tax revision or an off-path tax announcement, discretionary taxation and firms' investment choices at government reputation zero are sequentially optimal. Firms also optimize at every prescribed history. These choices and the stated beliefs support the selected plan. $\square$

B.7 Proof of Proposition 7

Proof. The proof holds for any fixed finite horizon T . Without profit shifting, investment $\mathcal{K}(w)$ solves $w\rho'(\mathcal{K}) = B\rho'(\bar{k} - \mathcal{K})$, where $B = 1 - \tau^*$, and profits earned in the host country are $S(w) = \rho(\mathcal{K}(w))$. Both functions are continuous and strictly increasing on $[0, 1]$, with $\mathcal{K}(0) = S(0) = 0$. Current revenue is at most $S(1)$, so discounted revenue over T periods is bounded by $M_T = S(1) \sum_{j=0}^{T-1} \beta^j$. Once the opportunistic type is revealed, backward induction gives confiscation and zero investment at every subsequent date.

I denote by $B_T(\delta)$ the supremum of the opportunistic type's discounted revenue over all sequentially rational plans lasting T periods, with common tax announcements and initial government reputation in $[0, \delta]$. Policies may depend on history, and future

government reputation need not remain below δ . I prove by induction on T that $B_T(\delta) \rightarrow 0$ as $\delta \downarrow 0$ and that date-1 investment converges uniformly to zero over these plans.

For $T = 1$, a tax of one is strictly optimal for the opportunistic type whenever capital is positive. Thus any equilibrium with positive investment and a tax announcement $\tau < 1$ has expected share of profits remaining after tax $w = p(1 - \tau) \leq p$, giving $B_1(\delta) \leq S(\delta)$. A tax announcement of one gives zero investment directly.

The induction hypothesis is that the claims hold for plans lasting $T - 1$ periods. At a date-1 tax announcement $\tau < 1$, the implementation probability is $u = p + (1 - p)m$, the posterior after implementation is $q = p/u$, and the expected after-tax return per unit of profit earned in the host country is $w = u(1 - \tau)$. If investment did not converge uniformly to zero, some sequence with $p \downarrow 0$ would have $w \geq \epsilon > 0$. Then $u \geq \epsilon$ and $1 - \tau \geq \epsilon$, so $m > 0$ at sufficiently small p . The opportunistic type must weakly prefer implementing the announced tax to confiscating profits, implying

$$\epsilon S(\epsilon) \leq (1 - \tau)S(w) \leq \beta B_{T-1}(p/\epsilon),$$

because $q \leq p/\epsilon$. The right side converges to zero, a contradiction. A tax announcement of one gives zero current investment. Current investment and revenue therefore vanish uniformly.

At a given history, the opportunistic type's discounted continuation revenue over the remaining $T - 1$ periods after implementing the date-1 announced tax is denoted by $W_{T-1}(q)$. For any $\eta \in (0, 1)$, its expected continuation revenue satisfies the bound below. If $q \leq \eta$, this value is at most $B_{T-1}(\eta)$. If $q > \eta$, Bayes' rule gives $u = p/q < p/\eta$ and $m \leq u/(1 - p)$. Hence

$$mW_{T-1}(q) \leq B_{T-1}(\eta) + \frac{p}{\eta(1 - p)}M_{T-1}.$$

Continuation revenue after a tax revision is zero. Taking $p \downarrow 0$ and then $\eta \downarrow 0$ proves that expected continuation revenue also vanishes uniformly. At $\tau = 1$, the implemented tax conveys no information and discounted continuation revenue is at most $\beta B_{T-1}(p)$. This completes the induction. Production and current revenue vanish because $zK^\alpha \rightarrow 0$ and $0 \leq \mathcal{R}(K, \tau) \leq \rho(K)$.

With profit shifting, the proof of Proposition 1 establishes that the after-tax return per unit of profit earned in the host country is at least $\bar{y}$ times the corresponding return on profit earned abroad at every implemented tax. Taking expectations preserves this inequality. The investment condition gives

$$K \geq \underline{K} > 0, \quad \bar{y}\rho'(\underline{K}) = \rho'(\bar{k} - \underline{K}).$$

This lower bound is independent of government reputation, implementation probabilities, and horizon length. Production in the host country is at least $z\bar{K}^\alpha > 0$. The bounds in both cases hold across all equilibria with common tax announcements. $\square$

C Numerical Methods

C.1 Numerical Solution

For the baseline model, the discretionary tax solves (11). For each candidate tax announcement, I solve the firm's investment problem and calculate the commitment type's revenue. I then choose the revenue-maximizing tax announcement subject to both types' announcement incentive compatibility constraints. The maximization includes boundary solutions and tax rates at which the direction of profit shifting changes or a tax-base constraint binds.

For the two-period model, I solve the implementation incentive compatibility conditions in Section 5.2 at each candidate tax announcement and maximize the commitment type's discounted revenue subject to both announcement incentive compatibility constraints. The maximization includes tax rates at which the opportunistic type's implementation probability reaches zero or one. For sustainable taxes, I maximize current revenue subject to (18) and (19). The finite-horizon calculation applies (22) backward from the terminal equilibrium.

C.2 Alternative Profit-Shifting Costs

The baseline places deductible profit-shifting costs in the country from which profits are transferred. I examine whether the optimal announced tax still declines with government reputation when those costs are incurred or deducted differently. Bilicka et al. (2024) and Altshuler et al. (2025) distinguish the costs of transferring profits from the costs of establishing a tax-planning structure. McClure (2023) discusses operational costs, audit exposure, and financial-reporting incentives. I compare the baseline with costs incurred in both countries, with and without deductibility.

When costs are incurred and deducted in both countries, the firm chooses its profit transfer to solve

$$\begin{aligned} \max_{\theta} \quad & (1 - \tau) \left[\rho(k) + \theta - \frac{\gamma\theta^2}{2\rho(k)} \right] \\ & + (1 - \tau^*) \left[\rho(\bar{k} - k) - \theta - \frac{\gamma\theta^2}{2\rho(\bar{k} - k)} \right], \end{aligned} \quad (37)$$

subject to nonnegative tax bases in both countries. When the constraints are slack, the optimal transfer is

$$\theta(k, \tau) = \frac{(\tau^* - \tau)\rho(k)\rho(\bar{k} - k)}{\gamma[(1 - \tau)\rho(\bar{k} - k) + (1 - \tau^*)\rho(k)]}. \quad (38)$$

The feasible interval is

$$\begin{aligned} \theta &\geq \max \left\{ \frac{\rho(k)(1 - \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 - \sqrt{1 + 2\gamma})}{\gamma} \right\}, \\ \theta &\leq \min \left\{ \frac{\rho(k)(1 + \sqrt{1 + 2\gamma})}{\gamma}, \frac{\rho(\bar{k} - k)(-1 + \sqrt{1 + 2\gamma})}{\gamma} \right\}. \end{aligned} \quad (39)$$

The objective is strictly concave in the transfer. The constrained optimum therefore equals the transfer in (38) when it is feasible and the nearest endpoint of the feasible interval otherwise. If operating profits are zero in either country, only a zero transfer is feasible.

With non-deductible costs in both countries, the firm chooses its profit transfer to maximize

$$\begin{aligned} &(1 - \tau)[\rho(k) + \theta] + (1 - \tau^*)[\rho(\bar{k} - k) - \theta] \\ &\quad - \frac{\gamma\theta^2}{2} \left[\frac{1}{\rho(k)} + \frac{1}{\rho(\bar{k} - k)} \right]. \end{aligned} \quad (40)$$

The unconstrained transfer is $(\tau^* - \tau)\rho(k)\rho(\bar{k} - k) / [\gamma(\rho(k) + \rho(\bar{k} - k))]$. The nonnegative tax bases restrict the transfer to $[-\rho(k), \rho(\bar{k} - k)]$. For both alternatives, I solve investment and discretionary taxation jointly and compare the commitment type's revenue across common tax announcements satisfying both types' announcement incentive compatibility constraints.

I choose γ in each cost specification to match the tax-base semi-elasticity at equal tax rates and equal operating profits, holding capital fixed. This semi-elasticity is the negative derivative of the log tax base in the host country with respect to its tax rate. It equals $1/[\gamma(1 - \tau^*)]$ under the baseline cost, $1/[2\gamma(1 - \tau^*)]$ when costs are deducted in both countries, and $1/(2\gamma)$ when costs in both countries are not deductible. The reference values are 1.26, the mean firm-specific semi-elasticity in [Kallen \(2023\)](#), and 1.50, the estimate for recent sample years in [Beer et al. \(2018\)](#). [Bilicka et al. \(2024\)](#) also report an aggregate short-run decline of 1.5 percent in reported profits after a one-percentage-point tax increase, holding productive capital and tax-planning capacity fixed. These estimates determine the response of reported profits to taxes used in the comparison, while production parameters remain those of the baseline economy.

TABLE 7: Optimal taxes with alternative profit-shifting costs

Costs incurred in	Tax deductible?	Tax-base semi-elasticity	Optimal announced tax (%)		
			$p = 0.1$	$p = 0.5$	$p = 0.9$
Both countries	Yes	1.26	31.5	29.3	27.2
Both countries	No	1.26	38.1	33.8	30.2
Country from which profits are shifted	Yes	1.26	30.6	29.3	27.9
Both countries	Yes	1.50	29.2	27.3	25.6
Both countries	No	1.50	34.3	30.9	28.0
Country from which profits are shifted	Yes	1.50	28.5	27.3	26.2

Notes: Taxes are in percent. The reference semi-elasticities are evaluated at equal tax rates and equal operating profits, holding capital fixed. Production parameters are the same across cost specifications. Tax bases must be nonnegative.

Table 7 reports the optimal one-period taxes for each cost specification. At both reference semi-elasticities, the optimal announced tax decreases across the three reported government reputation levels. Table 8 then holds the baseline cost specification fixed and varies its semi-elasticity to examine the size of the losses in first-period investment and output in the host country. Each loss is measured against the corresponding full-commitment allocation.

TABLE 8: First-period real activity relative to full commitment

Tax-base semi-elasticity	Investment loss (%)			Output loss (%)		
	$p = 0.1$	$p = 0.5$	$p = 0.9$	$p = 0.1$	$p = 0.5$	$p = 0.9$
Without profit shifting	93.91	52.44	3.51	84.22	38.77	2.33
Baseline	7.05	3.82	0.30	4.71	2.54	0.20
1.26	4.96	2.68	0.20	3.30	1.77	0.13
1.50	4.05	2.18	0.16	2.69	1.44	0.11

Notes: Allocations under the optimal common first-period tax announcement with $\beta = 0.6$. Investment and output refer to the host country, with losses measured relative to each economy's full-commitment allocation. The baseline cost specification has $\gamma = 1.25$ and tax-base semi-elasticity $1/[\gamma(1 - \tau^*)] = 0.941$ at equal tax rates and equal operating profits. For the rows labeled 1.26 and 1.50, γ is chosen to match the stated semi-elasticity. Production parameters are unchanged. In every reported allocation with profit shifting, the opportunistic type implements the first-period announced tax with certainty. Without profit shifting, it sometimes revises the announced tax at $p = 0.1$ and implements it with certainty at $p = 0.5$ and 0.9 .

C.3 Finite Changes in Profit-Shifting Costs

Table 2 compares two economies differing in a cost parameter set before the first tax announcement and held fixed in both periods. Each economy has its own discretionary tax, continuation revenues, and optimal first-period tax announcement and implemen-

tation probability. The full-commitment comparison uses the corresponding cost in each economy. It distinguishes the tax response under uncertain commitment from the response when the government must implement its announced rate.

Table 2 separates the investment response at the original announced tax from the response after the government chooses its optimal tax under the higher cost. In both columns, firms' reporting choices adjust to the new cost. The difference between the columns is the additional change in investment due to the higher optimal announced tax, expressed relative to the original investment level. The fixed-tax allocation assumes implementation of the original tax even though the opportunistic type would prefer to revise it under the higher cost.

To explain why the original tax announcement becomes harder to sustain, I hold that tax announcement and its implementation probability fixed while changing the cost from γ_0 to γ_1 . The initial economy's selected tax announcement and implementation probability are (τ_0, m_0) , with overall implementation probability $u_0 = p + (1 - p)m_0$ and posterior $q_0 = p/u_0$. For economy $i \in \{0, 1\}$, investment at the original announced tax is $\tilde{K}_i = K_{\gamma_i}(\tau_0, u_0)$ and the discretionary tax is τ_i^O . The current gain from a tax revision and the future revenue loss are

$$\begin{aligned} G_i &= \mathcal{R}_{\gamma_i}(\tilde{K}_i, \tau_i^O) - \mathcal{R}_{\gamma_i}(\tilde{K}_i, \tau_0), \\ L_i &= V_{O, \gamma_i}(q_0) - R_{0, \gamma_i}. \end{aligned} \quad (41)$$

The change in the net gain from a tax revision is

$$\Delta D = \Delta G - \beta \Delta L, \quad \beta \Delta L = \beta \Delta V_O(q_0) - \beta \Delta R_0. \quad (42)$$

Investment at the fixed tax announcement and implementation probability, discretionary taxation, and terminal policies all adjust to the new cost. At $p = 0.5$, the increase from $\gamma = 1.25$ to 1.50 raises G by 0.01240 and βL by 0.00130 in model revenue units. Both increases are positive at each reported prior, but the current gain rises more, weakening the incentive to implement the original announced tax rate. The government responds by announcing a higher tax that both types implement (Table 2).

I repeat the cost comparison in the baseline economy for five cost values (0.5, 0.75, 1, 1.25, and 1.5), three discount factors (0.1, 0.6, and 0.9), and seven priors (0.001, 0.01, 0.1, 0.25, 0.5, 0.75, and 0.9). All values of the profit-shifting cost parameter satisfy (6). I compare consecutive cost values and the change from 0.5 to 1.25. Investment falls and expected current revenue rises in each comparison.